MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DECODER, DIGITAL, KEY CODE AHA

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic decoder module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.371 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance and requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be AHA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/65-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

3.10 Functional tests. All modules shall be subjected to functional tests which conform to the requirements stated in 3.10.1 through 3.10.3.

3.10.1 Test patterns. A set of unique input patterns shall be generated capable of detecting all nonredundant failures in the module logic network. The test patterns shall be generated by assuming "stuck at H state" and "stuck at L state" failures in each of the basic logic functions and applying input patterns to detect these failures. The unique test patterns necessary to satisfy these requirements shall be submitted with the qualification samples. If the tests of tables I and XIX satisfy or exceed these unique test patterns, no additional tests shall be added; otherwise, the needed tests shall be added to table XIX. "Stuck at H state" indicates a failure resulting in a steady high-level output. "Stuck at L state" indicates a failure resulting in a steady low-level output. Prior to any change in module design which would necessitate a different set of test patterns to detect a "stuck at H state" or a "stuck at L state", the vendor shall be required to submit to the procuring activity and the Standard Electronic Module Program Quality Assurance Activity the proposed new set of test patterns for approval.

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3.10.2 Input requirements. The following input voltage levels shall be used for test patterns:

Low level - Input voltage shall be 0.8 V ±1 percent.
High level - Input voltage shall be 2.0 V ±1 percent.

3.10.3 Output requirements. The output requirements for the test patterns shall be as follows:

Low level - Output voltage shall be 0 V minimum, 0.5 V maximum.
High level - Output voltage shall be 2.5 V minimum, V_{CC} maximum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

* 4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

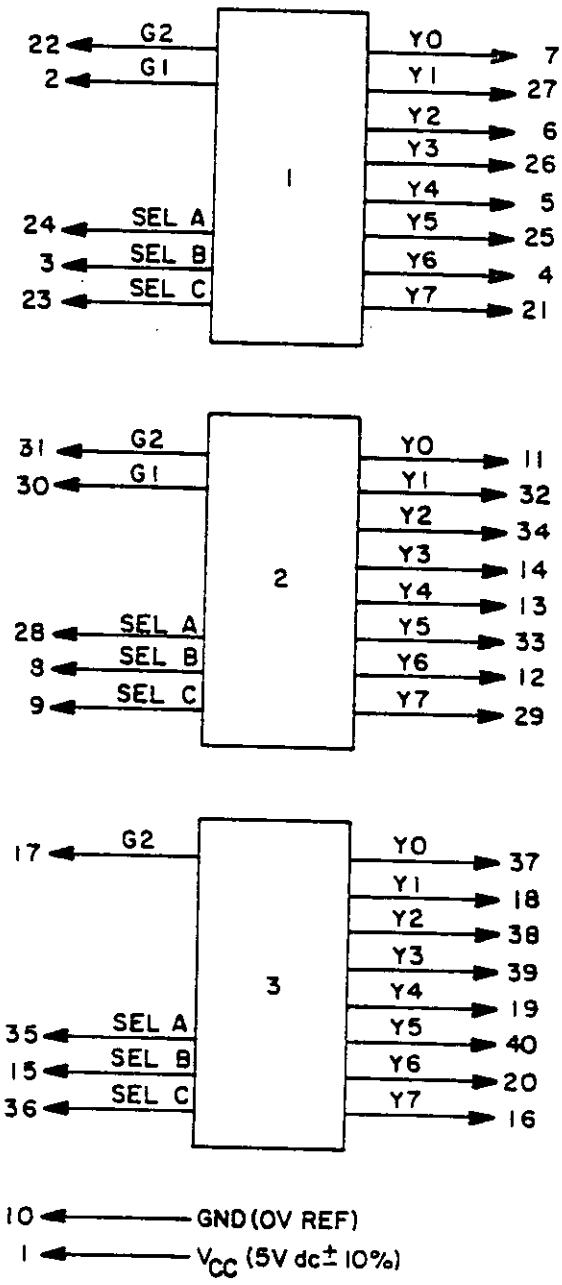
4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through U and W shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition U, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through U and W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through W shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

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NOTES:
1. G1, G2 = Enable inputs.
2. SEL A,B,C = Select inputs.
3. Y0-Y7 = Outputs.

* FIGURE 1. AHA module function diagram, decoder.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B24	V_{OUT} (low loaded)	IV	S	0.1	1.0	0.1	1.0	0.1	1.0	0.1	1.0	V
C1,C6,C11	Input breakdown current	V	S	0	2	0	2	0	2	0	2	mA
C2-C5,C7-C10, C12-C14	Input breakdown current	V	S	0	1	0	1	0	1	0	1	mA
D1-D24	Output short circuit current	VI	X	-40	-100	-40	-100	-40	-100	-40	-100	mA
E1-E24	Output leakage current	VII	X	0	1	0	1	0	1	0	1	mA
F1-F14	Input clamp diode voltage	VIII	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
G1-G24	Output voltage, low level	IX	X	0	500	0	500	0	500	0	500	mV
H1-H24	Output voltage, high level	X	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
J1,J6,J11	Input current, high level	XI	X	0	100	0	100	0	100	0	100	µA
J2-J5, J7-J10, J12-J14	Input current, high level	XI	X	0	50	0	50	0	50	0	50	µA
K1,K6,K11	Input current, low level	XII	X	-0.4	-4.4	-0.4	-4.4	-0.4	-4.4	-0.4	-4.4	mA
K2-K5, K7-K10, K12-K14	Input current, low level	XII	X	-0.4	-2.2	-0.4	-2.2	-0.4	-2.2	-0.4	-2.2	mA
L	Power supply current	XIII	X	50	222	50	222	50	222	50	222	mA
M1-M24	Transition time (t_{TLH}), C_L = 50 pF	XIV	S	2	15	2	15	2	18	2	18	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M25-M48	Transition time (t_{TLH}), C_L = 150 pF	XIV	Q	6	29	6	29	6	30	6	30	ns
N1-N24	Transition time (t_{THL}), C_L = 50 pF	XIV	S	1	7	1	7	1	8	1	8	ns
N25-N48	Transition time (t_{THL}), C_L = 150 pF	XIV	Q	3	17	3	17	3	17	3	17	ns
P1-P16	Propagation delay time (t_{PLH}), C_L = 50 pF	XIV	S	2	15	2	15	2	17	2	17	ns
P17-P32	Propagation delay time (t_{PLH}), C_L = 150 pF	XIV	Q	5	20	5	20	5	22	5	22	ns
P33-P56	Propagation delay time (t_{PLH}), C_L = 50 pF	XIV	S	2	15	2	15	2	17	2	17	ns
P57-P80	Propagation delay time (t_{PLH}), C_L = 150 pF	XIV	Q	4	20	4	20	4	22	4	22	ns
P81-P84, P89-P92, P97-P100, P105-P108, P113-P116, P121-P124, P129-P132, P137-P140, P145-P148	Propagation delay time (t_{PLH}), C_L = 50 pF	XIV	S	2	11	2	11	2	13	2	13	ns
P85-P88, P93-P96, P101-P104, P109-P112, P117-P120, P125-P128, P133-P136, P141-P144, P149-P152	Propagation delay time (t_{PLH}), C_L = 50 pF	XIV	S	2	15	2	15	2	17	2	17	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P153-P156, P161-P164, P169-P172, P177-P180, P185-P188, P193-P196, P201-P204, P209-P212, P217-P220	Propagation delay time (t_{PLH}), C_L = 150 pF	XIV	Q	3	17	3	17	3	19	3	19	ns
P157-P160, P165-P168, P173-P176, P181-P184, P189-P192, P197-P200, P205-P208, P213-P216, P221-P224	Propagation delay time (t_{PLH}), C_L = 150 pF	XIV	Q	4	20	4	20	4	22	4	22	ns
Q1-Q16	Propagation delay time (t_{PHL}), C_L = 50 pF	XIV	S	2	15	2	15	2	17	2	17	ns
Q17-Q32	Propagation delay time (t_{PHL}), C_L = 150 pF	XIV	Q	5	20	5	20	5	22	5	22	ns
Q33-Q56	Propagation delay time (t_{PHL}), C_L = 50 pF	XIV	S	2	15	2	15	2	17	2	17	ns
Q57-Q80	Propagation delay time (t_{PHL}), C_L = 150 pF	XIV	Q	5	20	5	20	5	22	5	22	ns
Q81-Q84, Q89-Q92, Q97-Q100, Q105-Q108, Q113-Q116, Q121-Q124, Q129-Q132, Q137-Q140, Q145-Q148	Propagation delay time (t_{PHL}), C_L = 50 pF	XIV	S	2	15	2	15	2	17	2	17	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q85-Q88, Q93-Q96, Q101-Q104, Q109-Q112, Q117-Q120, Q125-Q128, Q133-Q136, Q141-Q144, Q149-Q152	Propagation delay time (t_{PHL}), C_L = 50 pF	XIV	S	2	16	2	16	2	18	2	18	ns
Q153-Q156, Q161-Q164, Q169-Q172, Q177-Q180, Q185-Q188, Q193-Q196, Q201-Q204, Q209-Q212, Q217-Q220	Propagation delay time (t_{PHL}), C_L = 150 pF	XIV	Q	5	20	5	20	5	22	5	22	ns
Q157-Q160, Q165-Q168, Q173-Q176, Q181-Q184, Q189-Q192, Q197-Q200, Q205-Q208, Q213-Q216, Q221-Q224	Propagation delay time (t_{PHL}), C_L = 150 pF	XIV	Q	5	21	5	21	5	23	5	23	ns
R1-R24	Output crosstalk, low level	XV	Q	0	500	0	500	0	500	0	500	mV
S1-S24	Output crosstalk, high level	XV	Q	0	400	0	400	0	400	0	400	mV
T	Power supply transient amplitude	XVI	Q	0	400	0	400	0	400	0	400	mV
U	Power supply filter capacitance	XVII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
V1,V6,V11	Input capacitance	XVIII	Q	5	30	5	30	5	30	5	30	pF
V2-V5, V7-V10, V12-V14	Input capacitance	XVIII	Q	5	15	5	15	5	15	5	15	pF
W1-W40	Functional	XIX	X		Test vectors must produce output vectors as specified in the test table.							

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number MIL-M-28787/65-1 supersedes part number M28787/65. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation	1,221 mW (outputs not loaded)
Supply voltage	5.0 V dc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2 V minimum, V_{CC} maximum
Input current, low level	
A, B, C, G1 inputs	
V_{IL} = 0.5 V	-2 mA maximum
V_{IL} = 0.0 V	-2.2 mA maximum
Input current, low level	
G2 input	
V_{IL} = 0.5 V	-4 mA maximum
V_{IL} = 0.0 V	-4.4 mA maximum
Input current, high level	
A, B, C, G1 inputs	
V_{IH} = 2.5 V	50 µA maximum
V_{IH} = 5.5 V	1 mA maximum
Input current, high level	
G2 input	
V_{IH} = 2.5 V	100 µA maximum
V_{IH} = 5.5 V	2 mA maximum
Output voltage	
Low level (I_{OL} = 20 mA)	0.5 V maximum
High level (I_{OH} = -1 mA)	2.5 V maximum
Output drive current maximum capability	
Low level (V_{OL} = 0.5 V)	20 mA
High level (V_{OH} = 2.5 V)	-1 mA
Propagation delay time (T_{PHL}), C_L = 50 pF	
G1 to output	2 ns minimum, 17 ns maximum
G2 to output	2 ns minimum, 17 ns maximum
Select to output (2 gate delay)	2 ns minimum, 17 ns maximum
Select to output (3 gate delay)	2 ns minimum, 18 ns maximum

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Propagation delay time (T_{PHL}), C_L = 150 pF)

G1 to output	5 ns minimum, 22 ns maximum
G2 to output	5 ns minimum, 22 ns maximum
Select to output (2 gate delay)	5 ns minimum, 22 ns maximum
Select to output (3 gate delay)	5 ns minimum, 23 ns maximum

Propagation delay time (T_{PLH}), C_L = 50 pF)

G1 to output	2 ns minimum, 17 ns maximum
G2 to output	2 ns minimum, 17 ns maximum
Select to output (2 gate delay)	2 ns minimum, 13 ns maximum
Select to output (3 gate delay)	2 ns minimum, 17 ns maximum

Propagation delay time (T_{PLH}), C_L = 150 pF)

G1 to output	5 ns minimum, 22 ns maximum
G2 to output	4 ns minimum, 22 ns maximum
Select to output (2 gate delay)	3 ns minimum, 19 ns maximum
Select to output (3 gate delay)	4 ns minimum, 22 ns maximum

* 6.5 Module function. The module contains three independent 3-to-8 line decoders with separate enables.

6.5.1 Input to output relationship. Each decoder has the input to output relationship of table II for positive logic.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

TABLE II. Input to output relationship. 1/-4/

INPUTS					OUTPUTS							
ENABLE		SELECT										
G1	G2	C	B	A	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7
X	H	X	X	X	H	H	H	H	H	H	H	H
L	X	X	X	X	H	H	H	H	H	H	H	H
H	L	L	L	L	L	H	H	H	H	H	H	H
H	L	L	L	H	H	L	H	H	H	H	H	H
H	L	L	H	L	H	H	L	H	H	H	H	H
H	L	L	H	H	H	H	H	L	H	H	H	H
H	L	H	L	L	H	H	H	H	L	H	H	H
H	L	H	L	H	H	H	H	H	H	L	H	H
H	L	H	H	L	H	H	H	H	H	H	L	H
H	L	H	H	H	H	H	H	H	H	H	H	L

1/ H = High level.
2/ L = Low level.
3/ X = Irrelevant.
4/ Decoder circuit 3 has no G1 enable.

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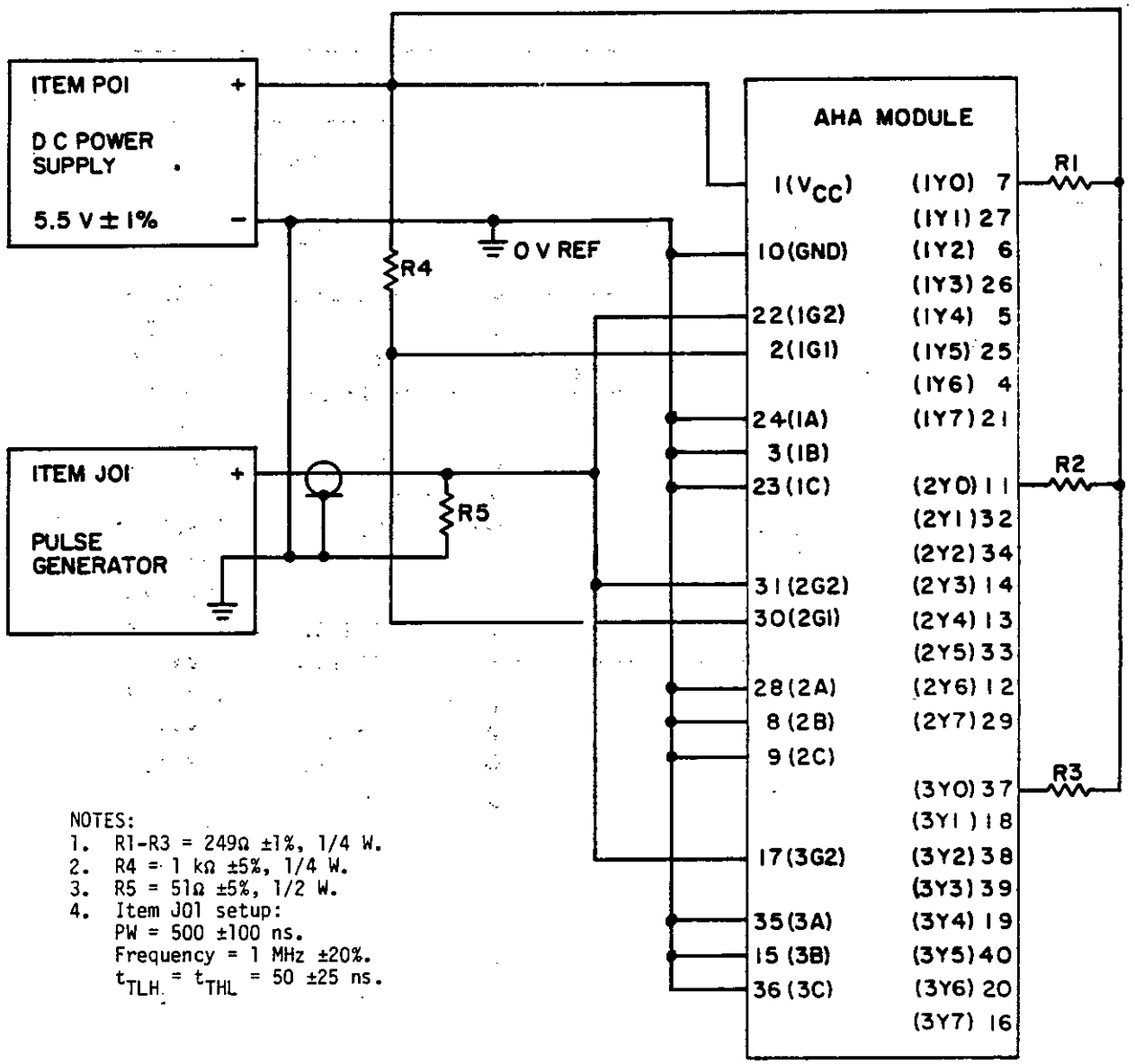
6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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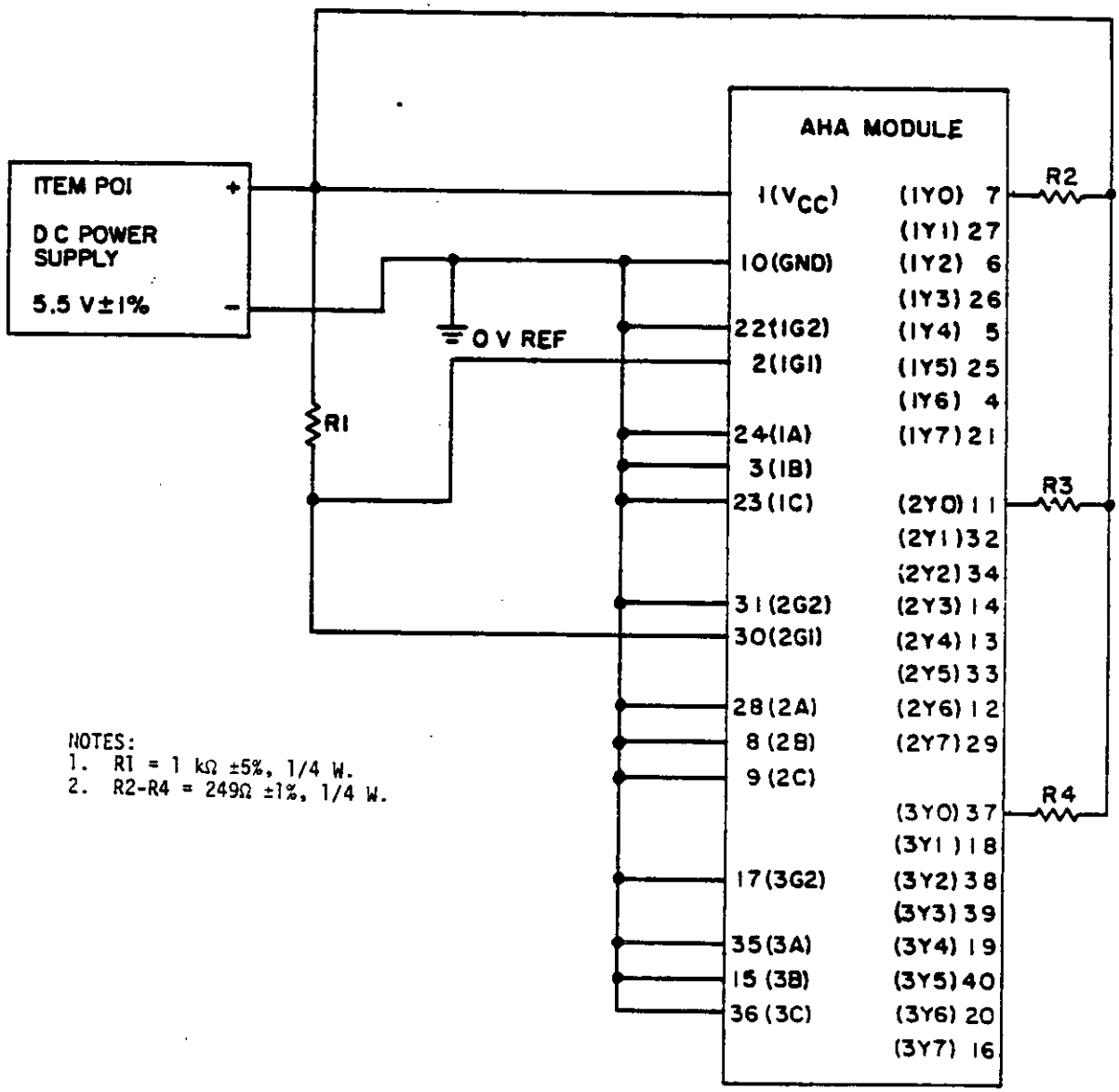
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NOTES:
1. R1-R3 = 249Ω ±1%, 1/4 W.
2. R4 = 1 kΩ ±5%, 1/4 W.
3. R5 = 51Ω ±5%, 1/2 W.
4. Item J01 setup:
   PW = 500 ±100 ns.
   Frequency = 1 MHz ±20%.
   $t_{TLH} = t_{THL}$ = 50 ±25 ns.

*FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = 1 kΩ ±5%, 1/4 W.
2. R2-R4 = 249Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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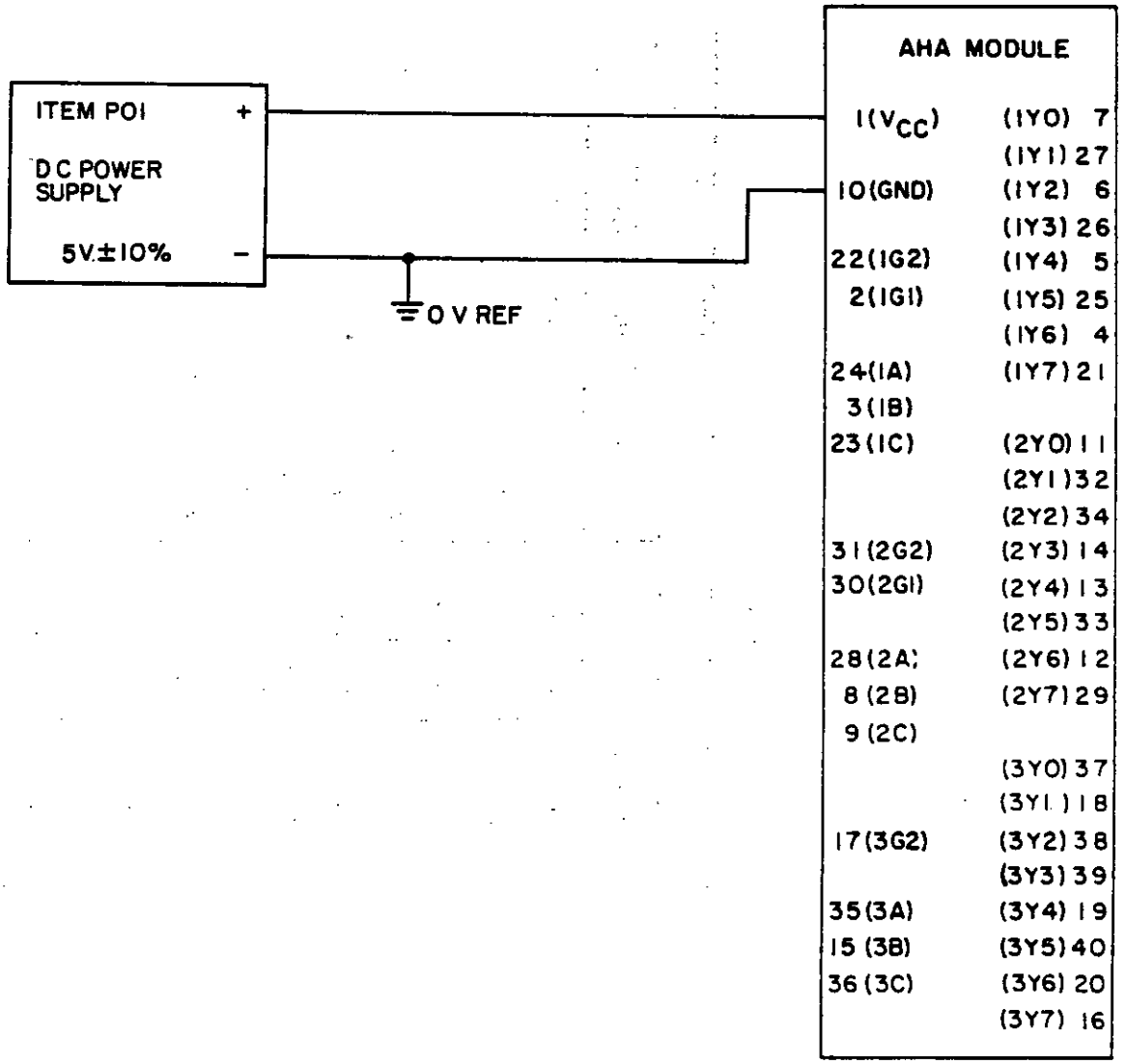


FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	All pins	Frame	10	∞	MΩ	5

TABLE IV. V_{OUT} (low-loaded) test.

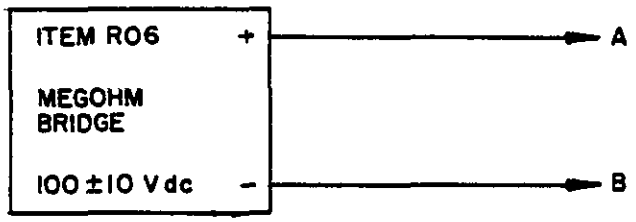
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
B1	1Y0	V_{OUT} (low loaded)	7	3,22,23,24	2	0.1	1.0	V	6
B2	1Y1	"	27	3,22,23	2,24	"	"	"	"
B3	1Y2	"	6	22,23,24	2,3	"	"	"	"
B4	1Y3	"	26	22,23	2,3,24	"	"	"	"
B5	1Y4	"	5	3,22,24	2,23	"	"	"	"
B6	1Y5	"	25	3,22	2,23,24	"	"	"	"
B7	1Y6	"	4	22,24	2,3,23	"	"	"	"
B8	1Y7	"	21	22	2,3,23,24	"	"	"	"
B9	2Y0	"	11	8,9,28,31	30	"	"	"	"
B10	2Y1	"	32	8,9,31	28,30	"	"	"	"
B11	2Y2	"	34	9,28,31	8,30	"	"	"	"
B12	2Y3	"	14	9,31	8,28,30	"	"	"	"

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TABLE IV. V_{OUT} (low-loaded) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
B13	2Y4	V_{OUT} (low loaded)	13	8,28,31	9,30	0.1	1.0	V	6
B14	2Y5	"	33	8,31	9,28,30	"	"	"	"
B15	2Y6	"	12	28,31	8,9,30	"	"	"	"
B16	2Y7	"	29	31	8,9,28,30	"	"	"	"
B17	3Y0	"	37	15,17,35, 36	Open	"	"	"	"
B18	3Y1	"	18	15,17,36	35	"	"	"	"
B19	3Y2	"	38	17,35,36	15	"	"	"	"
B20	3Y3	"	39	17,36	15,35	"	"	"	"
B21	3Y4	"	19	15,17,35	36	"	"	"	"
B22	3Y5	"	40	15,17	35,36	"	"	"	"
B23	3Y6	"	20	17,35	15,36	"	"	"	"
B24	3Y7	"	16	17	15,35,36	"	"	"	"

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WARNING:

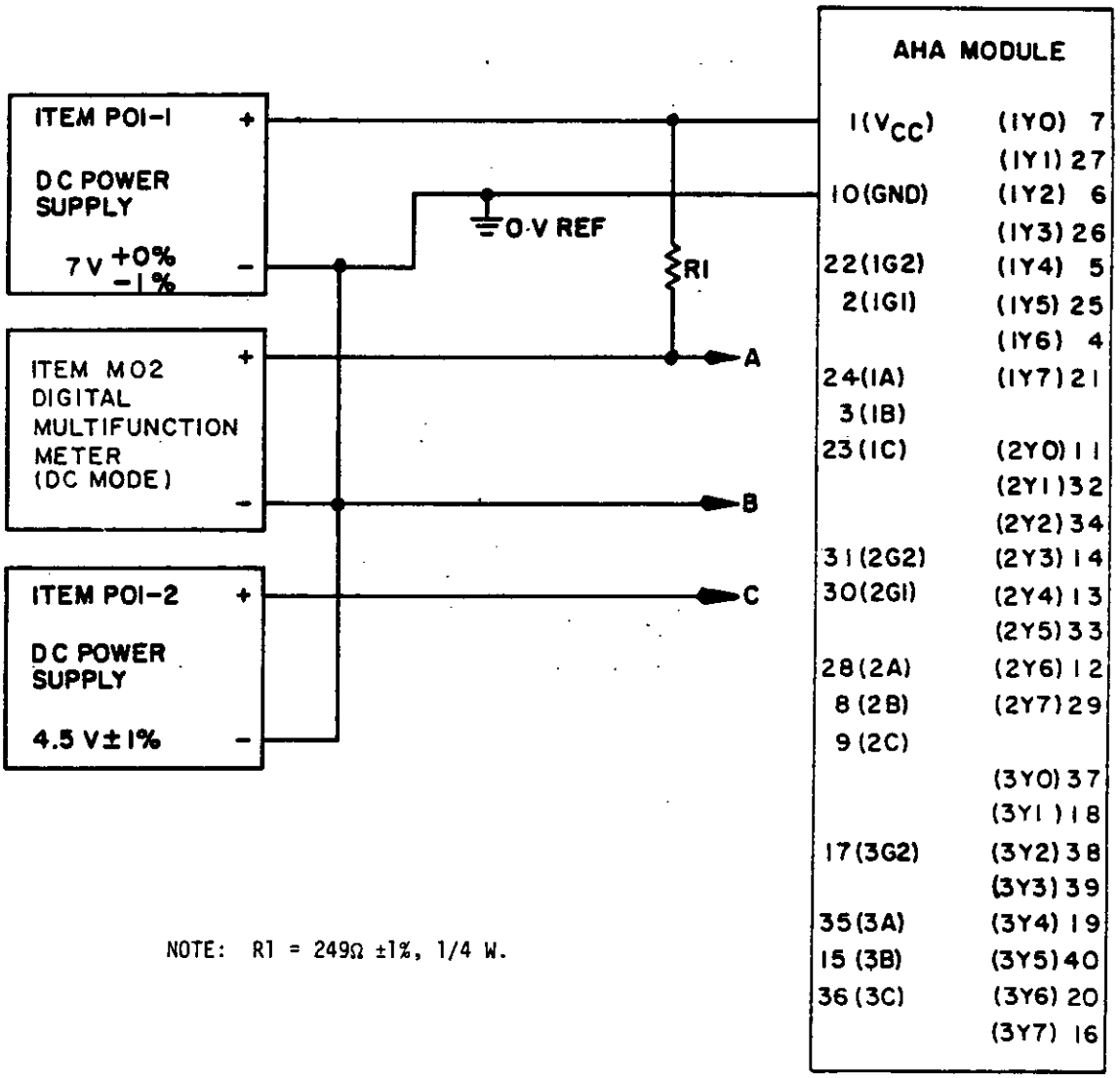
When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

AHA MODULE	
1(V_{CC})	(1Y0) 7
	(1Y1) 27
10(GND)	(1Y2) 6
	(1Y3) 26
22(1G2)	(1Y4) 5
2(1G1)	(1Y5) 25
	(1Y6) 4
24(1A)	(1Y7) 21
3(1B)	
23(1C)	(2Y0) 11
	(2Y1) 32
	(2Y2) 34
31(2G2)	(2Y3) 14
30(2G1)	(2Y4) 13
	(2Y5) 33
28(2A)	(2Y6) 12
8(2B)	(2Y7) 29
9(2C)	
	(3Y0) 37
	(3Y1) 18
17(3G2)	(3Y2) 38
	(3Y3) 39
35(3A)	(3Y4) 19
15(3B)	(3Y5) 40
36(3C)	(3Y6) 20
	(3Y7) 16

FIGURE 5. Isolation test circuit for table III.

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NOTE: R1 = 249Ω ±1%, 1/4 W.

* FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

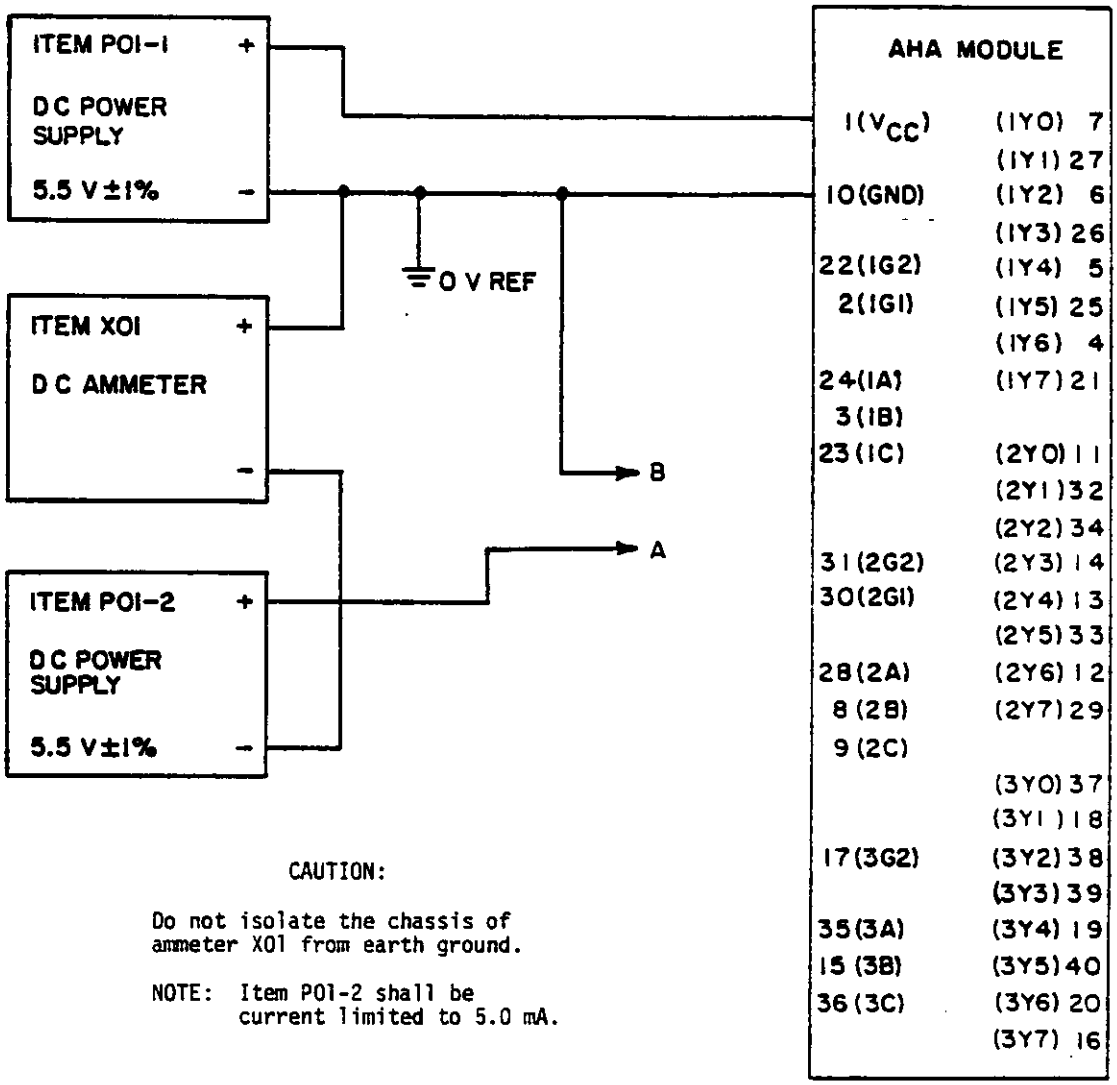
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	1G2	Input breakdown current	22	2,24,3,23, 31,30,28,8, 9,17,35,15, 36	0	2	mA	7
C2	1G1	"	2	22,24,3,23, 31,30,28,8, 9,17,15,35, 36	"	1	"	"
C3	1A	"	24	2,22,3,23, 31,30,28,8, 9,17,15,35, 36	"	"	"	"
C4	1B	"	3	2,22,24,23, 31,30,28,8, 9,17,35,15, 36	"	"	"	"
C5	1C	"	23	2,22,24,3, 31,30,28,8, 9,17,35,15, 36	"	"	"	"
C6	2G2		31	2,22,24,3, 23,30,28,8, 9,17,35,15, 36	"	2	"	"
C7	2G1	"	30	2,22,24,3, 23,31,28,8, 9,17,35,15, 36	"	1	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition		Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C8	2A	Input breakdown current	28	2,22,24,3, 31,30,8,9, 17,35,15,23, 36	0	1	mA	7
C9	2B	"	8	2,22,24,3, 23,31,30,28, 9,17,35,15, 36	"	"	"	"
C10	2C	"	9	2,22,24,23, 31,30,28,8, 17,35,15,36, 3	"	"	"	"
C11	3G2	"	17	2,22,24,3, 23,31,30,28, 8,9,35,15, 36	"	2	"	"
C12	3A	"	35	2,22,24,3, 23,31,30,28, 8,9,17,15,36	"	1	"	"
C13	3B	"	15	2,22,24,3, 23,31,30,28, 8,9,17,35,36	"	"	"	"
C14	3C	"	36	2,22,24,3, 23,31,30,28, 8,9,17,15,35	"	"	"	"

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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

NOTE: Item POI-2 shall be current limited to 5.0 mA.

* FIGURE 7. Input breakdown current test circuit for table V.

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TABLE VI. Output short circuit current test.

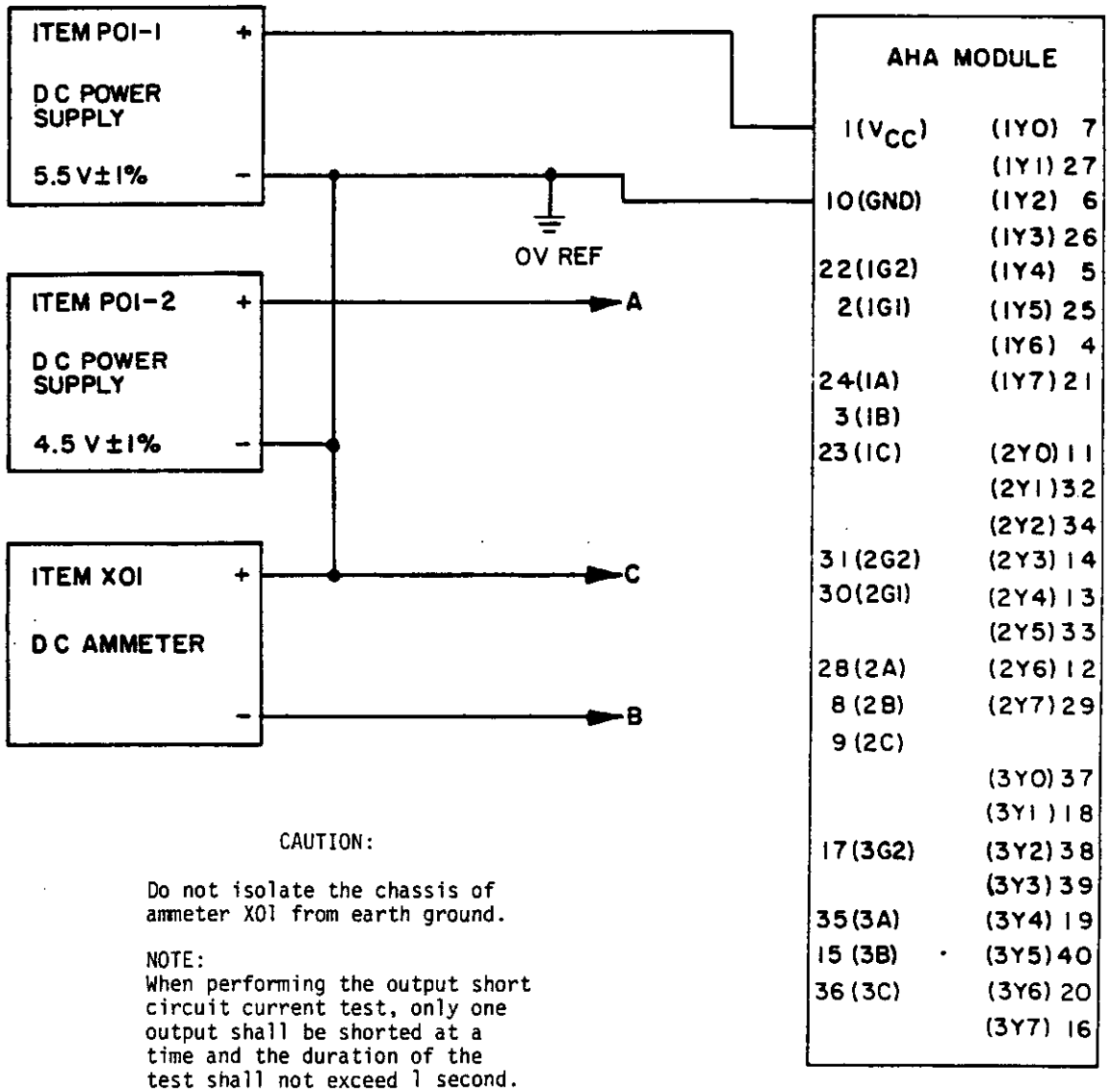
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
D1	1Y0	Output short circuit current	17,22,31	7	2,3,8,9, 15,23,24, 28,30,35, 36	-40	-100	mA	8
D2	1Y1	"	"	27	"	"	"	"	"
D3	1Y2	"	"	6	"	"	"	"	"
D4	1Y3	"	"	26	"	"	"	"	"
D5	1Y4	"	"	5	"	"	"	"	"
D6	1Y5	"	"	25	"	"	"	"	"
D7	1Y6	"	"	4	"	"	"	"	"
D8	1Y7	"	"	21	"	"	"	"	"
D9	2Y0	"	"	11	"	"	"	"	"
D10	2Y1	"	"	32	"	"	"	"	"
D11	2Y2	"	"	34	"	"	"	"	"
D12	2Y3	"	"	14	"	"	"	"	"
D13	2Y4	"	"	13	"	"	"	"	"
D14	2Y5	"	"	33	"	"	"	"	"
D15	2Y6	"	"	12	"	"	"	"	"
D16	2Y7	"	"	29	"	"	"	"	"

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TABLE VI. Output short circuit current test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
D17	3Y0	Output short circuit current	17,22,31	37	2,3,8,9, 15,23,24, 28,30,35, 36	-40	-100	mA	8
D18	3Y1	"	"	18	"	"	"	"	"
D19	3Y2	"	"	38	"	"	"	"	"
D20	3Y3	"	"	39	"	"	"	"	"
D21	3Y4	"	"	19	"	"	"	"	"
D22	3Y5	"	"	40	"	"	"	"	"
D23	3Y6	"	"	20	"	"	"	"	"
D24	3Y7	"	"	16	"	"	"	"	"

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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

NOTE:
When performing the output short circuit current test, only one output shall be shorted at a time and the duration of the test shall not exceed 1 second.

*FIGURE 8. Output short circuit current test circuit for table VI.

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TABLE VII. Output leakage current test.

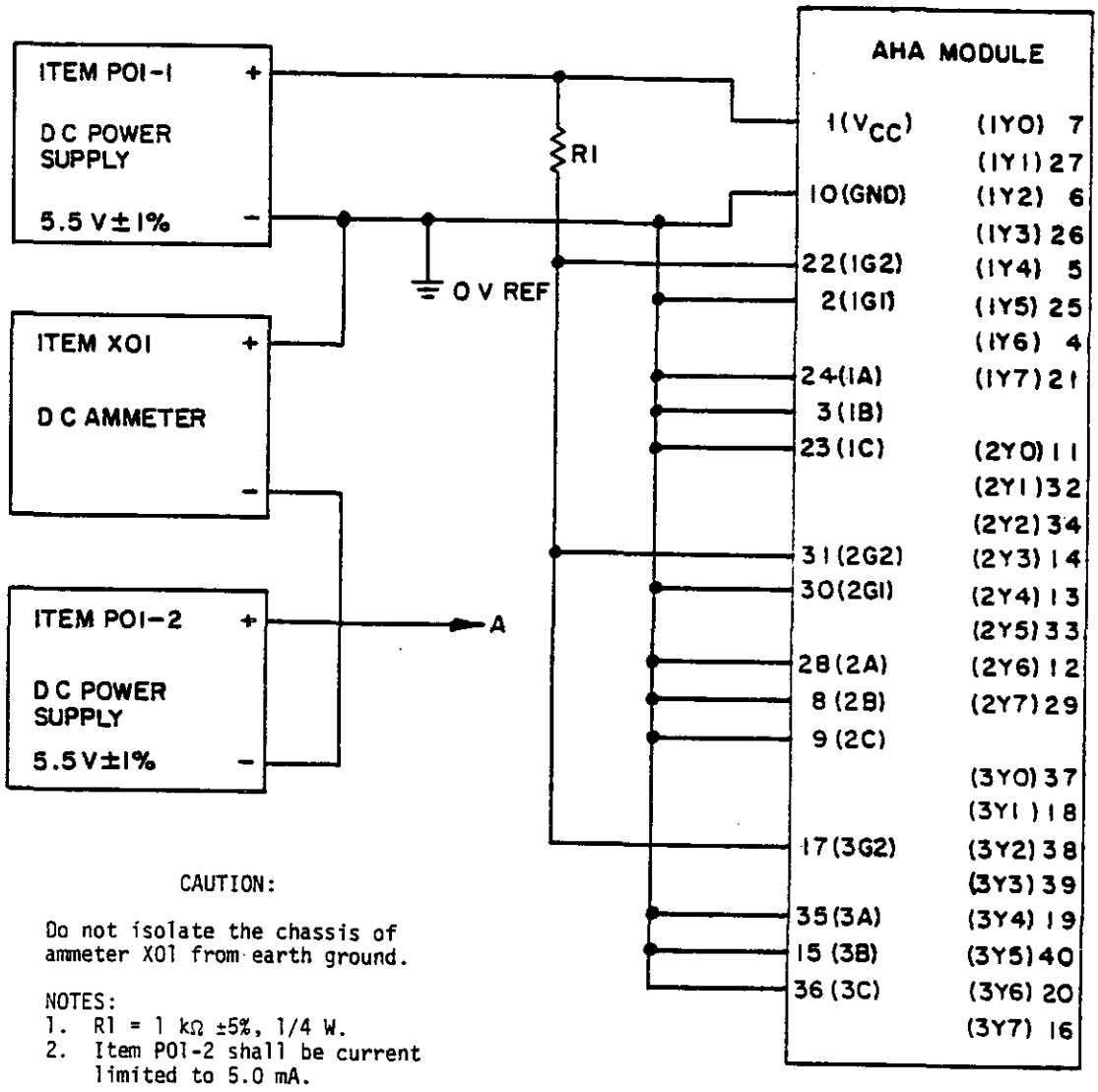
Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
E1	1Y0	Output leakage current	7	0	1	mA	9
E2	1Y1	"	27	"	"	"	"
E3	1Y2	"	6	"	"	"	"
E4	1Y3	"	26	"	"	"	"
E5	1Y4	"	5	"	"	"	"
E6	1Y5	"	25	"	"	"	"
E7	1Y6	"	4	"	"	"	"
E8	1Y7	"	21	"	"	"	"
E9	2Y0	"	11	"	"	"	"
E10	2Y1	"	32	"	"	"	"
E11	2Y2	"	34	"	"	"	"
E12	2Y3	"	14	"	"	"	"
E13	2Y4	"	13	"	"	"	"
E14	2Y5	"	33	"	"	"	"
E15	2Y6	"	12	"	"	"	"
E16	2Y7	"	29	"	"	"	"
E17	3Y0	"	37	"	"	"	"
E18	3Y1	"	18	"	"	"	"
E19	3Y2	"	38	"	"	"	"
E20	3Y3	"	39	"	"	"	"

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TABLE VII. Output leakage current test - Continued.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
E21	3Y4	Output leakage current	19	0	1	mA	9
E22	3Y5	"	40	"	"	"	"
E23	3Y6	"	20	"	"	"	"
E24	3Y7	"	16	"	"	"	"

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. R1 = 1 kΩ ±5%, 1/4 W.
2. Item PO1-2 shall be current limited to 5.0 mA.

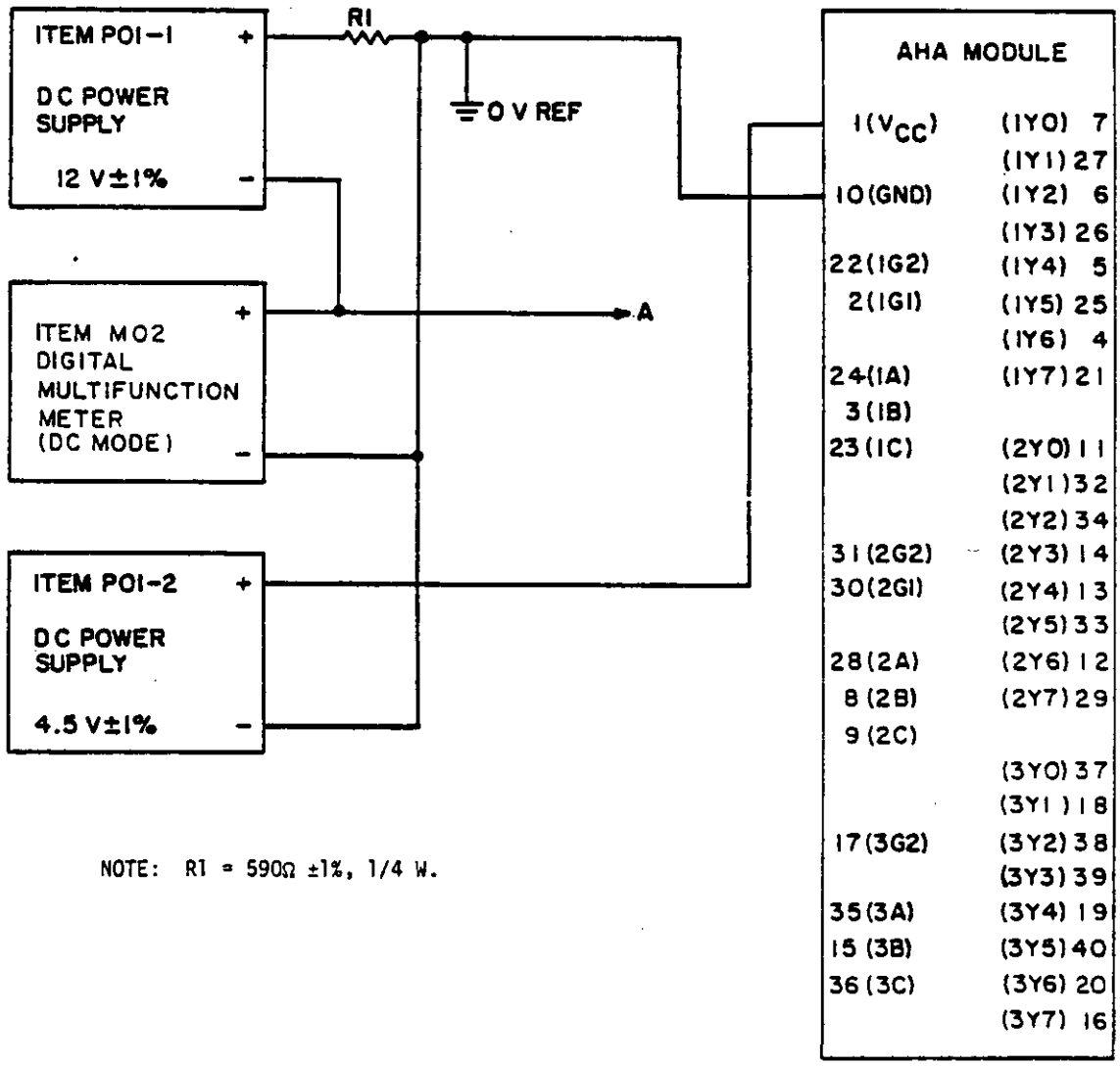
FIGURE 9. Output leakage current test circuit for table VII.

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TABLE VIII. Input clamp diode voltage test.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
F1	1G2	Input clamp diode voltage	22	-0.4	-1.2	V	10
F2	1G1	"	2	"	"	"	"
F3	1A	"	24	"	"	"	"
F4	1B	"	3	"	"	"	"
F5	1C	"	23	"	"	"	"
F6	2G2	"	31	"	"	"	"
F7	2G1	"	30	"	"	"	"
F8	2A	"	28	"	"	"	"
F9	2B	"	8	"	"	"	"
F10	2C	"	9	"	"	"	"
F11	3G2	"	17	"	"	"	"
F12	3A	"	35	"	"	"	"
F13	3B	"	15	"	"	"	"
F14	3C	"	36	"	"	"	"

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NOTE: R1 = 590Ω ±1%, 1/4 W.

* FIGURE 10. Input clamp diode voltage test circuit for table VIII.

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TABLE IX. Output voltage (low-level) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G1	1Y0	Output voltage, low level	2	22,24,23,3	7	0	500	mV	11
G2	1Y1	"	2,24	22,23,3	27	"	"	"	"
G3	1Y2	"	2,3	22,24,23	6	"	"	"	"
G4	1Y3	"	2,3,24	22,23	26	"	"	"	"
G5	1Y4	"	2,23	22,3,24	5	"	"	"	"
G6	1Y5	"	2,23,24	22,3	25	"	"	"	"
G7	1Y6	"	2,3,23	22,24	4	"	"	"	"
G8	1Y7	"	2,3,23,24	22	21	"	"	"	"
G9	2Y0	"	30	31,8,9,28	11	"	"	"	"
G10	2Y1	"	30,28	31,8,9	32	"	"	"	"
G11	2Y2	"	30,8	31,9,28	34	"	"	"	"
G12	2Y3	"	30,8,28	31,9	14	"	"	"	"
G13	2Y4	"	30,9	31,8,28	13	"	"	"	"
G14	2Y5	"	30,9,28	31,8	33	"	"	"	"
G15	2Y6	"	30,9,8	31,28	12	"	"	"	"
G16	2Y7	"	30,8,9,28	31	29	"	"	"	"
G17	3Y0	"	Open	17,15,35, 36	37	"	"	"	"
G18	3Y1	"	35	17,15,36	18	"	"	"	"
G19	3Y2	"	15	17,35,36	38	"	"	"	"
G20	3Y3	"	15,35	17,36	39	"	"	"	"

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TABLE IX. Output voltage (low-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G21	3Y4	Output voltage, low level	36	17,15,35	19	0	500	mV	11
G22	3Y5	"	35,36	17,15	40	"	"	"	"
G23	3Y6	"	15,36	17,35	20	"	"	"	"
G24	3Y7	"	15,35,36	17	16	"	"	"	"

TABLE X. Output voltage (high-level) test.

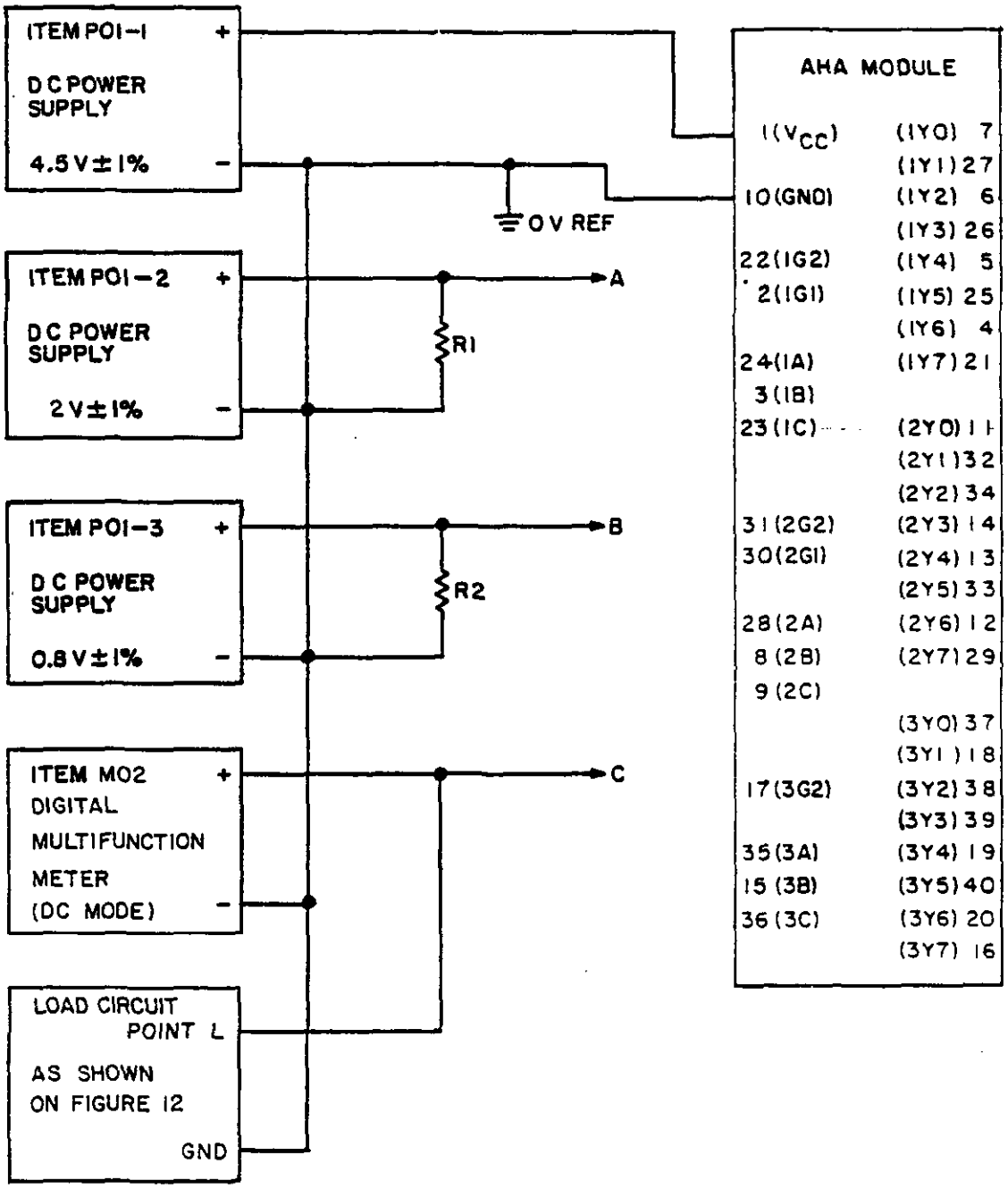
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H1	1Y0	Output voltage, high level	22	Open	7	2,5	4.5	V	11
H2	1Y1	"	"	"	27	"	"	"	"
H3	1Y2	"	"	"	6	"	"	"	"
H4	1Y3	"	"	"	26	"	"	"	"
H5	1Y4	"	"	"	5	"	"	"	"
H6	1Y5	"	"	"	25	"	"	"	"
H7	1Y6	"	"	"	4	"	"	"	"
H8	1Y7	"	"	"	21	"	"	"	"
H9	2Y0	"	31	"	11	"	"	"	"

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TABLE X. Output voltage (high-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H10	2Y1	Output voltage, high level	31	Open	32	2.5	4.5	V	11
H11	2Y2	"	"	"	34	"	"	"	"
H12	2Y3	"	"	"	14	"	"	"	"
H13	2Y4	"	"	"	13	"	"	"	"
H14	2Y5	"	"	"	33	"	"	"	"
H15	2Y6	"	"	"	12	"	"	"	"
H16	2Y7	"	"	"	29	"	"	"	"
H17	3Y0	"	17	"	37	"	"	"	"
H18	3Y1	"	"	"	18	"	"	"	"
H19	3Y2	"	"	"	38	"	"	"	"
H20	3Y3	"	"	"	39	"	"	"	"
H21	3Y4	"	"	"	19	"	"	"	"
H22	3Y5	"	"	"	40	"	"	"	"
H23	3Y6	"	"	"	20	"	"	"	"
H24	3Y7	"	"	"	16	"	"	"	"

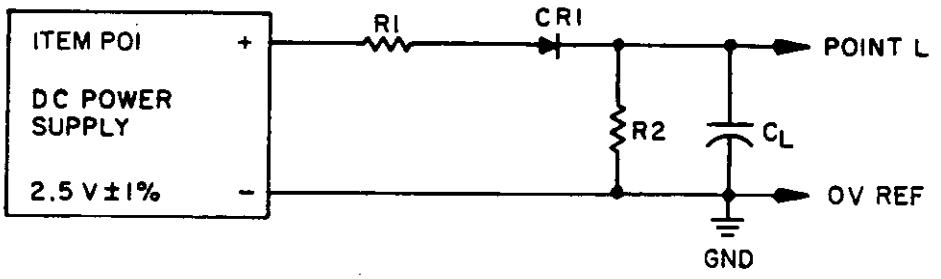
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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

* FIGURE 11. Output voltage (low-level and high-level) test circuit for tables IX and X.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 20 mA ±5% out of point L with point L held at 0.5 V.
3. R2 = 2.49 kΩ ±1%, 1/8 W.
4. Test type S for test conditions M1-M24, N1-N24, P1-P16, P33-P56, P81-P152, Q1-Q16, Q33-Q56, and Q81-Q152. C_L = 50 ±5 pF (including probe and fixture capacitance). Test type Q for test conditions M25-M48, N25-N48, P17-P32, P57-P80, P153-P224, Q17-Q32, Q57-Q80, and Q153-Q224. C_L = 150 ±5 pF (including probe and fixture capacitance) except for test conditions R and S where C_L = 50 ±5 pF (including probe and fixture capacitance).
5. Capacitor C_L is not required for dc electrical test conditions G1-G24 and H1-H24.

* FIGURE 12. AHA load circuit.

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TABLE XI. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	1G2 Input current, high level	22	2,5,8,9,15, 17,23,24,28, 30,31,35,36	0	100	µA	13
J2	1G1 "	2	3,8,9,15,17, 22,23,24,28, 30,31,35,36	"	50	"	"
J3	1A "	24	2,3,8,9,15, 17,22,23,28, 30,31,35,36	"	"	"	"
J4	1B "	3	2,8,9,15,17, 22,23,24,28, 30,31,35,36	"	"	"	"
J5	1C "	23	2,3,8,9,15, 17,22,24,28, 30,31,35,36	"	"	"	"
J6	2G2 "	31	2,3,8,9,15, 17,22,23,24, 28,30,35,36	"	100	"	"
J7	2G1 "	30	2,3,8,9,15, 17,22,23,24, 28,31,35,36	"	50	"	"
J8	2A "	28	2,3,8,9,15, 17,22,23,24, 30,31,35,36	"	"	"	"
J9	2B "	8	2,3,9,15,17, 22,23,24,28, 30,31,35,36	"	"	"	"

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TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J10	2C Input current, high level	9	2,3,8,15,17, 22,23,24,28, 30,31,35,36	0	50	µA	13
J11	3G2 "	17	2,3,8,9,15, 22,23,24,28, 30,31,35,36	"	100	"	"
J12	3A "	35	2,3,8,9,15, 17,22,23,24, 28,30,31,36	"	50	"	"
J13	3B "	15	2,3,8,9,17, 22,23,24,28, 30,31,35,36	"	"	"	"
J14	3C "	36	2,3,8,9,15, 17,22,23,24, 28,30,31,35	"	"	"	"

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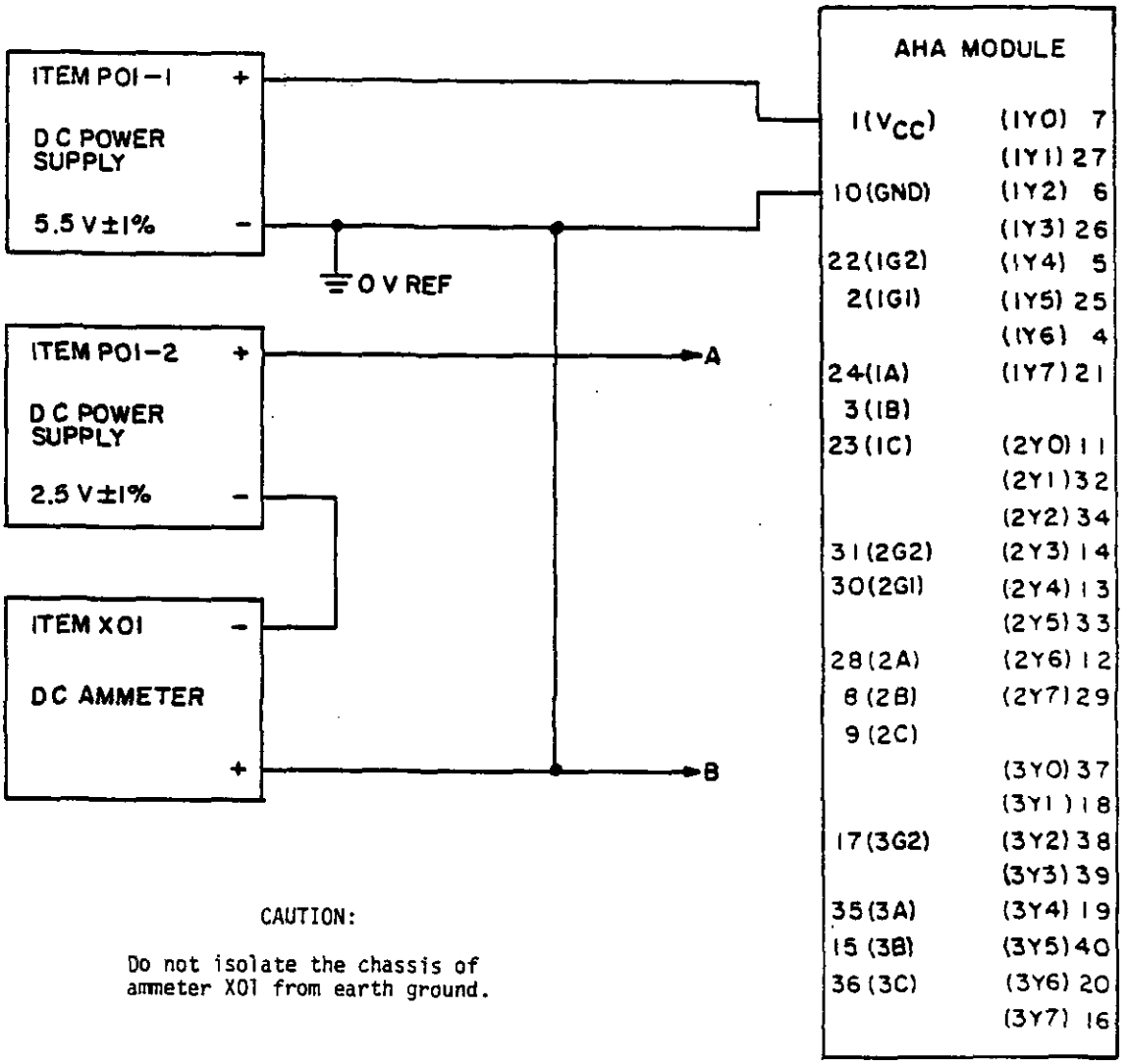


FIGURE 13. Input current (high-level) test circuit for table XI.

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TABLE XII. Input current (low-level) test.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
K1	1G2	Input current, low level	22	-0.4	-4.4	mA	14
K2	1G1	"	2	"	-2.2	"	"
K3	1A	"	24	"	"	"	"
K4	1B	"	3	"	"	"	"
K5	1C	"	23	"	"	"	"
K6	2G2	"	31	"	-4.4	"	"
K7	2G1	"	30	"	-2.2	"	"
K8	2A	"	28	"	"	"	"
K9	2B	"	8	"	"	"	"
K10	2C	"	9	"	"	"	"
K11	3G2	"	17	"	-4.4	"	"
K12	3A	"	35	"	-2.2	"	"
K13	3B	"	15	"	"	"	"
K14	3C	"	36	"	"	"	"

TABLE XIII. Power supply current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
L	Power supply current	17,22,31	50	222	mA	15

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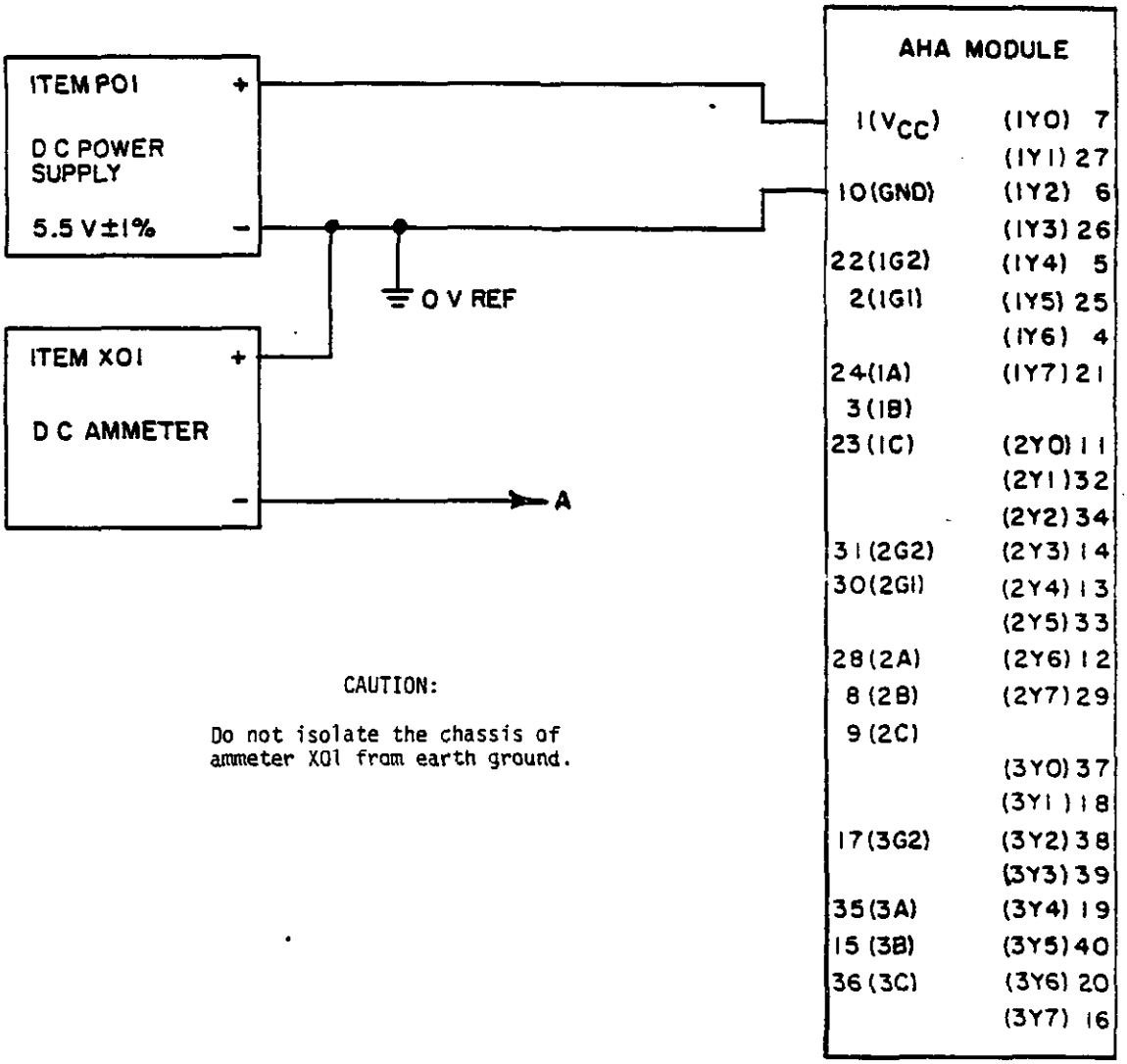
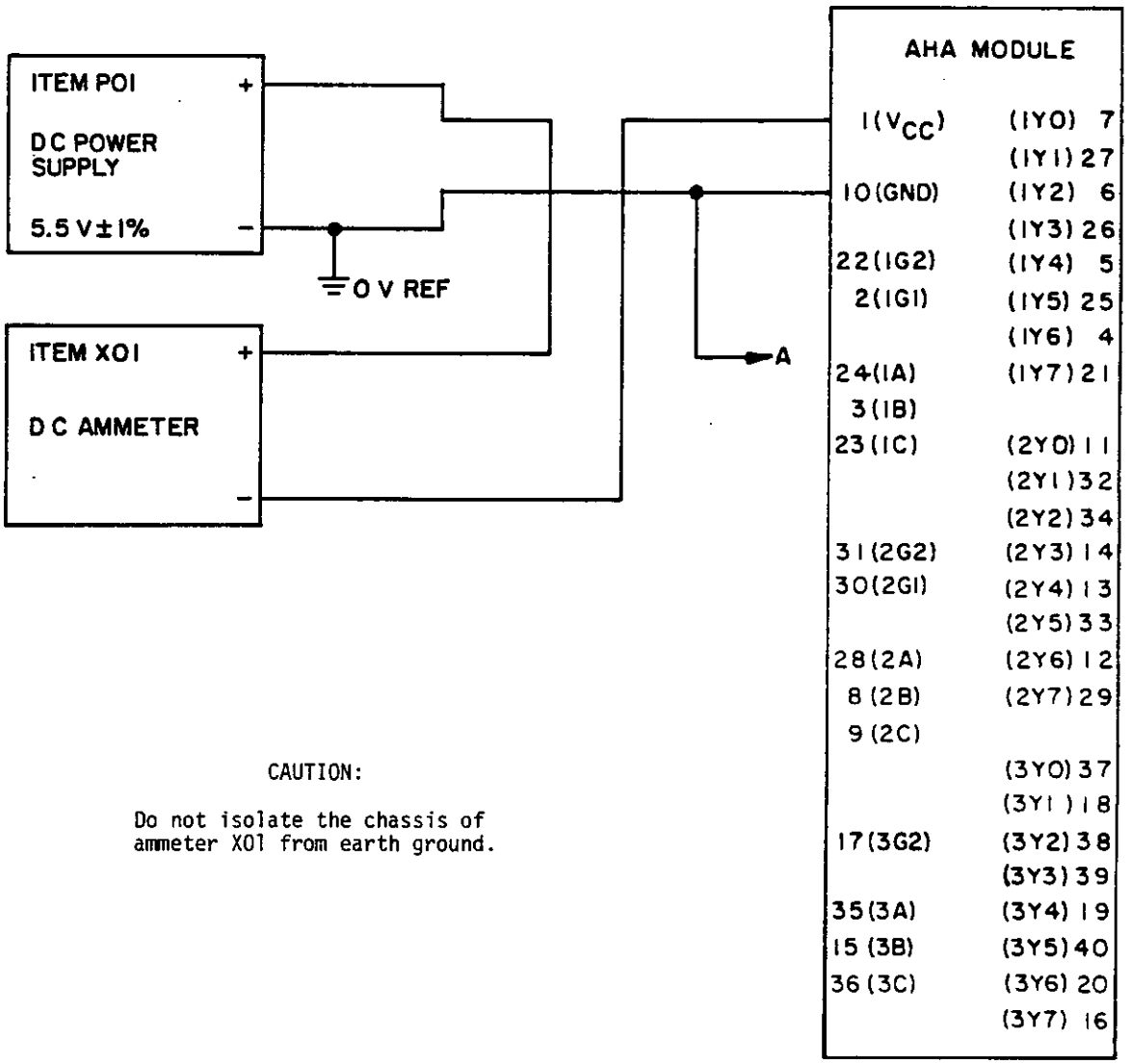


FIGURE 14. Input current (low-level) test circuit for table XII.

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 15. Power supply current test circuit for table XIII.

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	1Y0 Transition time (t_{TLH}), C_L = 50 pF	2	3,22, 24	23	7	2	15	ns	16,17
M2	1Y1 "	2,24	3,22	"	27	"	"	"	"
M3	1Y2 "	2,3	22,24	"	6	"	"	"	"
M4	1Y3 "	2,3,24	22	"	26	"	"	"	"
M5	1Y4 "	2	3,22, 24	"	5	"	"	"	"
M6	1Y5 "	2,24	3,22	"	25	"	"	"	"
M7	1Y6 "	2,3	22,24	"	4	"	"	"	"
M8	1Y7 "	2,3,24	22	"	21	"	"	"	"
M9	2Y0 "	30	8,28, 31	9	11	"	"	"	"
M10	2Y1 "	28,30	8,31	"	32	"	"	"	"
M11	2Y2 "	8,30	28,31	"	34	"	"	"	"
M12	2Y3 "	8,28, 30	31	"	14	"	"	"	"
M13	2Y4 "	30	8,28, 31	"	13	"	"	"	"
M14	2Y5 "	28,30	8,31	"	33	"	"	"	"
M15	2Y6 "	8,30	28,31	"	12	"	"	"	"
M16	2Y7 "	8,28, 30	31	"	29	"	"	"	"
M17	3Y0 "	Open	15,17, 35	36	37	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M18	3Y1	Transition time (t_{TLH}), C_L = 50 pF	35	15,17	36	18	2	15	ns	16,17
M19	3Y2	"	15	17,35	"	38	"	"	"	"
M20	3Y3	"	15,35	17	"	39	"	"	"	"
M21	3Y4	"	Open	15,17, 35	"	19	"	"	"	"
M22	3Y5	"	35	15,17	"	40	"	"	"	"
M23	3Y6	"	15	17,35	"	20	"	"	"	"
M24	3Y7	"	15,35	17	"	16	"	"	"	"
M25	1Y0	Transition time (t_{TLH}), C_L = 150 pF	2	3,22, 24	23	7	6	29	"	"
M26	1Y1	"	2,24	3,22	"	27	"	"	"	"
M27	1Y2	"	2,3	22,24	"	6	"	"	"	"
M28	1Y3	"	2,3,24	22	"	26	"	"	"	"
M29	1Y4	"	2	3,22, 24	"	5	"	"	"	"
M30	1Y5	"	2,24	3,22	"	25	"	"	"	"
M31	1Y6	"	2,3	22,24	"	4	"	"	"	"
M32	1Y7	"	2,3,24	22	"	21	"	"	"	"
M33	2Y0	"	30	8,28, 31	9	11	"	"	"	"
M34	2Y1	"	28,30	8,31	9	32	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M35	2Y2	Transition time (t_{TLH}), C_L = 150 pF	8,30	28,31	9	34	6	29	ns	16,17
M36	2Y3	"	8,28, 30	31	"	14	"	"	"	"
M37	2Y4	"	30	8,28,31	"	13	"	"	"	"
M38	2Y5	"	28,30	8,31	"	33	"	"	"	"
M39	2Y6	"	8,30	28,31	"	12	"	"	"	"
M40	2Y7	"	8,28, 30	31	"	29	"	"	"	"
M41	3Y0	"	Open	15,17, 35	36	37	"	"	"	"
M42	3Y1	"	35	15,17	"	18	"	"	"	"
M43	3Y2	"	15	17,35	"	38	"	"	"	"
M44	3Y3	"	15,35	17	"	39	"	"	"	"
M45	3Y4	"	Open	15,17, 35	"	19	"	"	"	"
M46	3Y5	"	35	15,17	"	40	"	"	"	"
M47	3Y6	"	15	17,35	"	20	"	"	"	"
M48	3Y7	"	15,35	17	"	16	"	"	"	"
N1	1Y0	Transition time (t_{THL}), C_L = 50 pF	2	3,22, 24	23	7	1	7	"	"
N2	1Y1	"	2,24	3,22	"	27	"	"	"	"
N3	1Y2	"	2,3	22,24	"	6	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N4	1Y3 Transition time (t_{THL}), C_L = 50 pF	2,3,24	22	23	26	1	7	ns	16,17
N5	1Y4 "	2	3,22,24	"	5	"	"	"	"
N6	1Y5 "	2,24	3,22	"	25	"	"	"	"
N7	1Y6 "	2,3	22,24	"	4	"	"	"	"
N8	1Y7 "	2,3,24	22	"	21	"	"	"	"
N9	2Y0 "	30	8,28, 31	9	11	"	"	"	"
N10	2Y1 "	28,30	8,31	"	32	"	"	"	"
N11	2Y2 "	8,30	28,31	"	34	"	"	"	"
N12	2Y3 "	8,28, 30	31	"	14	"	"	"	"
N13	2Y4 "	30	8,28, 31	"	13	"	"	"	"
N14	2Y5 "	28,30	8,31	"	33	"	"	"	"
N15	2Y6 "	8,30	28,31	"	12	"	"	"	"
N16	2Y7 "	8,28, 30	31	"	29	"	"	"	"
N17	3Y0 "	Open	15,17, 35	36	37	"	"	"	"
N18	3Y1 "	35	15,17	"	18	"	"	"	"
N19	3Y2 "	15	17,35	"	38	"	"	"	"
N20	3Y3 "	15,35	17	"	39	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N21	3Y4	Transition time (t_{THL}), C_L = 50 pF	Open	15,17,35	36	19	1	7	ns	16,17
N22	3Y5	"	35	15,17	"	40	"	"	"	"
N23	3Y6	"	15	17,35	"	20	"	"	"	"
N24	3Y7	"	15,35	17	"	16	"	"	"	"
N25	1Y0	Transition time (t_{THL}), C_L = 150 pF	2	3,22, 24	23	7	3	17	"	"
N26	1Y1	"	2,24	3,22	"	27	"	"	"	"
N27	1Y2	"	2,3	22,24	"	6	"	"	"	"
N28	1Y3	"	2,3,24	22	"	26	"	"	"	"
N29	1Y4	"	2	3,22, 24	"	5	"	"	"	"
N30	1Y5	"	2,24	3,22	"	25	"	"	"	"
N31	1Y6	"	2,3	22,24	"	4	"	"	"	"
N32	1Y7	"	2,3,24	22	"	21	"	"	"	"
N33	2Y0	"	30	8,28, 31	9	11	"	"	"	"
N34	2Y1	"	28,30	8,31	"	32	"	"	"	"
N35	2Y2	"	8,30	28,31	"	34	"	"	"	"
N36	2Y3	"	8,28,30	31	"	14	"	"	"	"
N37	2Y4	"	30	8,28,31	"	13	"	"	"	"
N38	2Y5	"	28,30	8,31	"	33	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N39	2Y6 Transition time (t_{THL}), C_L = 150 pF	8,30	28,31	9	12	3	17	ns	16,17
N40	2Y7 "	8,28,30	31	9	29	"	"	"	"
N41	3Y0 "	Open	15,17, 35	36	37	"	"	"	"
N42	3Y1 "	35	15,17	"	18	"	"	"	"
N43	3Y2 "	15	17,35	"	38	"	"	"	"
N44	3Y3 "	15,35	17	"	39	"	"	"	"
N45	3Y4 "	Open	15,17, 35	"	19	"	"	"	"
N46	3Y5 "	35	15,17	"	40	"	"	"	"
N47	3Y6 "	15	17,35	"	20	"	"	"	"
N48	3Y7 "	15,35	17	"	16	"	"	"	"
P1	1G1-1Y0 Propagation delay time (t_{PLH}), C_L = 50 pF	Open	3,22, 23,24	2	7	2	15	"	"
P2	1G1-1Y1 "	24	3,22, 23	"	27	"	"	"	"
P3	1G1-1Y2 "	3	22,23, 24	"	6	"	"	"	"
P4	1G1-1Y3 "	3,24	22,23	"	26	"	"	"	"
P5	1G1-1Y4 "	23	3,22, 24	"	5	"	"	"	"
P6	1G1-1Y5 "	23,24	3,22	"	25	"	"	"	"
P7	1G1-1Y6 "	3,23	22,24	"	4	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P8	1G1-1Y7	Propagation delay time (t_{PLH}), C_L = 50 pF	3,23, 24	22	2	21	2	15	ns	16,17
P9	2G1-2Y0	"	Open	8,9, 28,31	30	11	"	"	"	"
P10	2G1-2Y1	"	28	8,9,31	"	32	"	"	"	"
P11	2G1-2Y2	"	8	9,28, 31	"	34	"	"	"	"
P12	2G1-2Y3	"	8,28	9,31	"	14	"	"	"	"
P13	2G1-2Y4	"	9	8,28, 31	"	13	"	"	"	"
P14	2G1-2Y5	"	9,28	8,31	"	33	"	"	"	"
P15	2G1-2Y6	"	9,8	28,31	"	12	"	"	"	"
P16	2G1-2Y7	"	8,9,28	31	"	29	"	"	"	"
P17	1G1-1Y0	Propagation delay time (t_{PLH}), C_L = 150 pF	Open	3,22, 23,24	2	7	5	20	"	"
P18	1G1-1Y1	"	24	3,22, 23	"	27	"	"	"	"
P19	1G1-1Y2	"	3	22,23, 24	"	6	"	"	"	"
P20	1G1-1Y3	"	3,24	22,23	"	26	"	"	"	"
P21	1G1-1Y4	"	23	3,22, 24	"	5	"	"	"	"
P22	1G1-1Y5	"	23,24	3,22	"	25	"	"	"	"
P23	1G1-1Y6	"	3,23	22,24	"	4	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P24	1G1-1Y7	Propagation delay time (t_{PLH}), C_L = 150 pF	3,23, 24	22	2	21	5	20	ns	16,17
P25	2G1-2Y0	"	Open	8,9, 28,31	30	11	"	"	"	"
P26	2G1-2Y1	"	28	8,9,31	"	32	"	"	"	"
P27	2G1-2Y2	"	8	9,28, 31	"	34	"	"	"	"
P28	2G1-2Y3	"	8,28	9,31	"	14	"	"	"	"
P29	2G1-2Y4	"	9	8,28, 31	"	13	"	"	"	"
P30	2G1-2Y5	"	9,28	8,31	"	33	"	"	"	"
P31	2G1-2Y6	"	8,9	28,31	"	12	"	"	"	"
P32	2G1-2Y7	"	8,9,28	31	"	29	"	"	"	"
P33	1G2-1Y0	Propagation delay time (t_{PLH}), C_L = 50 pF	2	3,23, 24	22	7	2	15	"	"
P34	1G2-1Y1	"	2,24	3,23	"	27	"	"	"	"
P35	1G2-1Y2	"	2,3	23,24	"	6	"	"	"	"
P36	1G2-1Y3	"	2,3,24	23	"	26	"	"	"	"
P37	1G2-1Y4	"	2,23	3,24	"	5	"	"	"	"
P38	1G2-1Y5	"	2,23, 24	3	"	25	"	"	"	"
P39	1G2-1Y6	"	2,3,23	24	"	4	"	"	"	"
P40	1G2-1Y7	"	2,3, 23,24	Open	"	21	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P41	2G2-2Y0	Propagation delay time (t_{PLH}), C_L = 50 pF	30	8,9,28	31	11	2	15	ns	16,17
P42	2G2-2Y1	"	28,30	8,9	"	32	"	"	"	"
P43	2G2-2Y2	"	8,30	9,28	"	34	"	"	"	"
P44	2G2-2Y3	"	8,28, 30	9	"	14	"	"	"	"
P45	2G2-2Y4	"	9,30	8,28	"	13	"	"	"	"
P46	2G2-2Y5	"	9,28, 30	8	"	33	"	"	"	"
P47	2G2-2Y6	"	8,9,30	28	"	12	"	"	"	"
P48	2G2-2Y7	"	8,9, 28,30	Open	"	29	"	"	"	"
P49	3G2-3Y0	"	Open	15,35, 36	17	37	"	"	"	"
P50	3G2-3Y1	"	35	15,36	"	18	"	"	"	"
P51	3G2-3Y2	"	15	35,36	"	38	"	"	"	"
P52	3G2-3Y3	"	15,35	36	"	39	"	"	"	"
P53	3G2-3Y4	"	36	15,35	"	19	"	"	"	"
P54	3G2-3Y5	"	35,36	15	"	40	"	"	"	"
P55	3G2-3Y6	"	15,36	35	"	20	"	"	"	"
P56	3G2-3Y7	"	15,35, 36	Open	"	16	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P57	1G2-1Y0	Propagation delay time (t_{PLH}), C_L = 150 pF	2	3,23, 24	22	7	4	20	ns	16,17
P58	1G2-1Y1	"	2,24	3,23	"	27	"	"	"	"
P59	1G2-1Y2	"	2,3	23,24	"	6	"	"	"	"
P60	1G2-1Y3	"	2,3,24	23	"	26	"	"	"	"
P61	1G2-1Y4	"	2,23	3,24	"	5	"	"	"	"
P62	1G2-1Y5	"	2,23, 24	3	"	25	"	"	"	"
P63	1G2-1Y6	"	2,3,23	24	"	4	"	"	"	"
P64	1G2-1Y7	"	2,3, 23,24	Open	"	21	"	"	"	"
P65	2G2-2Y0	"	30	8,9,28	31	11	"	"	"	"
P66	2G2-2Y1	"	28,30	8,9	"	32	"	"	"	"
P67	2G2-2Y2	"	8,30	9,28	"	34	"	"	"	"
P68	2G2-2Y3	"	8,28, 30	9	"	14	"	"	"	"
P69	2G2-2Y4	"	9,30	8,28	"	13	"	"	"	"
P70	2G2-2Y5	"	9,28, 30	8	"	33	"	"	"	"
P71	2G2-2Y6	"	8,9,30	28	"	12	"	"	"	"
P72	2G2-2Y7	"	8,9, 28,30	Open	"	29	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P73	3G2-3Y0	Propagation delay time (t_{PLH}), C_L = 150 pF	Open	15,35, 36	17	37	4	20	ns	16,17
P74	3G2-3Y1	"	35	15,36	"	18	"	"	"	"
P75	3G2-3Y2	"	15	35,36	"	38	"	"	"	"
P76	3G2-3Y3	"	15,35	36	"	39	"	"	"	"
P77	3G2-3Y4	"	36	15,35	"	19	"	"	"	"
P78	3G2-3Y5	"	35,36	15	"	40	"	"	"	"
P79	3G2-3Y6	"	15,36	35	"	20	"	"	"	"
P80	3G2-3Y7	"	15,35, 36	Open	"	16	"	"	"	"
P81	1C-1Y0	Propagation delay time (t_{PLH}), C_L = 50 pF	2	3,22, 24	23	7	2	11	"	"
P82	1C-1Y1	"	2,24	3,22	"	27	"	"	"	"
P83	1C-1Y2	"	2,3	22,24	"	6	"	"	"	"
P84	1C-1Y3	"	2,3,24	22	"	26	"	"	"	"
P85	1C-1Y4	"	2	3,22, 24	"	5	"	15	"	"
P86	1C-1Y5	"	2,24	3,22	"	25	"	"	"	"
P87	1C-1Y6	"	2,3	22,24	"	4	"	"	"	"
P88	1C-1Y7	"	2,3,24	22	"	21	"	"	"	"
P89	1B-1Y0	"	2	22,23, 24	3	7	"	11	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P90	1B-1Y1	Propagation delay time (t_{PLH}), C_L = 50 pF	2,24	22,23	3	27	2	11	ns	16,17
P91	1B-1Y4	"	2,23	22,24	"	5	"	"	"	"
P92	1B-1Y5	"	2,23, 24	22	"	25	"	"	"	"
P93	1B-1Y2	"	2	22,23, 24	"	6	"	15	"	"
P94	1B-1Y3	"	2,24	22,23	"	26	"	"	"	"
P95	1B-1Y6	"	2,23	22,24	"	4	"	"	"	"
P96	1B-1Y7	"	2,23, 24	22	"	21	"	"	"	"
P97	1A-1Y0	"	2	3,22, 23	24	7	"	11	"	"
P98	1A-1Y2	"	2,3	22,23	"	6	"	"	"	"
P99	1A-1Y4	"	2,23	3,22	"	5	"	"	"	"
P100	1A-1Y6	"	2,3,23	22	"	4	"	"	"	"
P101	1A-1Y1	"	2	3,22, 23	"	27	"	15	"	"
P102	1A-1Y3	"	2,3	22,23	"	26	"	"	"	"
P103	1A-1Y5	"	2,23	3,22	"	25	"	"	"	"
P104	1A-1Y7	"	2,3,23	22	"	21	"	"	"	"
P105	2C-2Y0	"	30	8,28, 31	9	11	"	11	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P106	2C-2Y1	Propagation delay time (t_{PLH}), C_L = 50 pF	28,30	8,31	9	32	2	11	ns	16,17
P107	2C-2Y2	"	8,30	28,31	"	34	"	"	"	"
P108	2C-2Y3	"	8,28, 30	31	"	14	"	"	"	"
P109	2C-2Y4	"	30	8,28, 31	"	13	"	15	"	"
P110	2C-2Y5	"	28,30	8,31	"	33	"	"	"	"
P111	2C-2Y6	"	8,30	28,31	"	12	"	"	"	"
P112	2C-2Y7	"	8,28, 30	31	"	29	"	"	"	"
P113	2B-2Y0	"	30	9,28, 31	8	11	"	11	"	"
P114	2B-2Y1	"	28,30	9,31	"	32	"	"	"	"
P115	2B-2Y4	"	9,30	28,31	"	13	"	"	"	"
P116	2B-2Y5	"	9,28, 30	31	"	33	"	"	"	"
P117	2B-2Y2	"	30	9,28, 31	"	34	"	15	"	"
P118	2B-2Y3	"	28,30	9,31	"	14	"	"	"	"
P119	2B-2Y6	"	9,30	28,31	"	12	"	"	"	"
P120	2B-2Y7	"	9,28, 30	31	"	29	"	"	"	"
P121	2A-2Y0	"	30	8,9,31	28	11	"	11	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P122	2A-2Y2	Propagation delay time (t_{PLH}), C_L = 50 pF	8,30	9,31	28	34	2	11	ns	16,17
P123	2A-2Y4	"	9,30	8,31	"	13	"	"	"	"
P124	2A-2Y6	"	8,9,30	31	"	12	"	"	"	"
P125	2A-2Y1	"	30	8,9,31	"	32	"	15	"	"
P126	2A-2Y3	"	8,30	9,31	"	14	"	"	"	"
P127	2A-2Y5	"	9,30	8,31	"	33	"	"	"	"
P128	2A-2Y7	"	8,9,30	31	"	29	"	"	"	"
P129	3C-2Y0	"	Open	15,17, 35	36	37	"	11	"	"
P130	3C-3Y1	"	35	15,17	"	18	"	"	"	"
P131	3C-3Y2	"	15	17,35	"	38	"	"	"	"
P132	3C-3Y3	"	15,35	17	"	39	"	"	"	"
P133	3C-3Y4	"	Open	15,17, 35	"	19	"	15	"	"
P134	3C-3Y5	"	35	15,17	"	40	"	"	"	"
P135	3C-3Y6	"	15	17,35	"	20	"	"	"	"
P136	3C-3Y7	"	15,35	17	"	16	"	"	"	"
P137	3B-3Y0	"	Open	17,35, 36	15	37	"	11	"	"
P138	3B-3Y1	"	35	17,36	15	18	"	11	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P139	3B-3Y4	Propagation delay time (t_{PLH}), C_L = 50 pF	36	17,35	15	19	2	11	ns	16,17
P140	3B-3Y5	"	35,36	17	"	40	"	11	"	"
P141	3B-3Y2	"	Open	17,35, 36	"	38	"	15	"	"
P142	3B-3Y3	"	35	17,36	"	39	"	"	"	"
P143	3B-3Y6	"	36	17,35	"	20	"	"	"	"
P144	3B-3Y7	"	35,36	17	"	16	"	"	"	"
P145	3A-3Y0	"	Open	15,17, 36	35	37	"	11	"	"
P146	3A-3Y2	"	15	17,36	"	38	"	"	"	"
P147	3A-3Y4	"	36	15,17	"	19	"	"	"	"
P148	3A-3Y6	"	15,36	17	"	20	"	"	"	"
P149	3A-3Y1	"	Open	15,17, 36	"	18	"	15	"	"
P150	3A-3Y3	"	15	17,36	"	39	"	"	"	"
P151	3A-3Y5	"	36	15,17	"	40	"	"	"	"
P152	3A-3Y7	"	15,36	17	"	16	"	"	"	"
P153	1C-1Y0	Propagation delay time (t_{PLH}), C_L = 150 pF	2	3,22, 24	23	7	3	17	"	"
P154	1C-1Y1	"	2,24	3,22	"	27	"	"	"	"
P155	1C-1Y2	"	2,3	22,24	"	6	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P156	1C-1Y3	Propagation delay time (t_{PLH}), C_L = 150 pF	2,3,24	22	23	26	3	17	ns	16,17
P157	1C-1Y4	"	2	3,22, 24	"	5	4	20	"	"
P158	1C-1Y5	"	2,24	3,22	"	25	"	"	"	"
P159	1C-1Y6	"	2,3	22,24	"	4	"	"	"	"
P160	1C-1Y7	"	2,3,24	22	"	21	"	"	"	"
P161	1B-1Y0	"	2	22,23, 24	3	7	3	17	"	"
P162	1B-1Y1	"	2,24	22,23	"	27	"	"	"	"
P163	1B-1Y4	"	2,3	22,24	"	5	"	"	"	"
P164	1B-1Y5	"	2,3,24	22	"	25	"	"	"	"
P165	1B-1Y2	"	2	22,23, 24	"	6	4	20	"	"
P166	1B-1Y3	"	2,24	22,23	"	26	"	"	"	"
P167	1B-1Y6	"	2,3	22,24	"	4	"	"	"	"
P168	1B-1Y7	"	2,3,24	22	"	21	"	"	"	"
P169	1A-1Y0	"	2	3,22, 23	24	7	3	17	"	"
P170	1A-1Y2	"	2,3	22,23	"	6	"	"	"	"
P171	1A-1Y4	"	2,23	3,22	"	5	"	"	"	"
P172	1A-1Y6	"	2,3,23	22	"	4	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P173	1A-1Y1	Propagation delay time (t_{PLH}), C_L = 150 pF	2	3,22, 23	24	27	4	20	ns	16,17
P174	1A-1Y3	"	2,3	22,23	"	26	"	"	"	"
P175	1A-1Y5	"	2,23	3,22	"	25	"	"	"	"
P176	1A-1Y7	"	2,3,23	22	"	21	"	"	"	"
P177	2C-2Y0	"	30	8,28, 31	9	11	3	17	"	"
P178	2C-2Y1	"	28,30	8,31	"	32	"	"	"	"
P179	2C-2Y2	"	8,30	28,31	"	34	"	"	"	"
P180	2C-2Y3	"	8,28, 30	31	"	14	"	"	"	"
P181	2C-2Y4	"	30	8,28, 31	"	13	4	20	"	"
P182	2C-2Y5	"	28,30	8,31	"	33	"	"	"	"
P183	2C-2Y6	"	8,30	28,31	"	12	"	"	"	"
P184	2C-2Y7	"	8,28, 30	31	"	29	"	"	"	"
P185	2B-2Y0	"	30	9,28, 31	8	11	3	17	"	"
P186	2B-2Y1	"	28,30	9,31	"	32	"	"	"	"
P187	2B-2Y4	"	9,30	28,31	"	13	"	"	"	"
P188	2B-2Y5	"	9,28, 30	31	"	33	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P189	2B-2Y2	Propagation delay time (t_{PLH}), C_L = 150 pF	30	9,28, 31	8	34	4	20	ns	16,17
P190	2B-2Y3	"	28,30	9,31	"	14	"	"	"	"
P191	2B-2Y6	"	9,30	28,31	"	12	"	"	"	"
P192	2B-2Y7	"	9,28, 30	31	"	29	"	"	"	"
P193	2A-2Y0	"	30	8,9,31	28	11	3	17	"	"
P194	2A-2Y2	"	8,30	9,31	"	34	"	"	"	"
P195	2A-2Y4	"	9,30	8,31	"	13	"	"	"	"
P196	2A-2Y6	"	8,9,30	31	"	12	"	"	"	"
P197	2A-2Y1	"	30	8,9,31	"	32	4	20	"	"
P198	2A-2Y3	"	8,30	9,31	"	14	"	"	"	"
P199	2A-2Y5	"	9,30	8,31	"	33	"	"	"	"
P200	2A-2Y7	"	8,9,30	31	"	29	"	"	"	"
P201	3C-3Y0	"	Open	15,17, 35	36	37	3	17	"	"
P202	3C-3Y1	"	35	15,17	"	18	"	"	"	"
P203	3C-3Y2	"	15	17,35	"	38	"	"	"	"
P204	3C-3Y3	"	15,35	17	"	39	"	"	"	"
P205	3C-3Y4	"	Open	15,17, 35	"	19	4	20	"	"
P206	3C-3Y5	"	35	15,17	"	40	4	20	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P207	3C-3Y6	Propagation delay time (t_{PLH}), C_L = 150 pF	15	17,35	36	20	4	20	ns	16,17
P208	3C-3Y7	"	15,35	17	36	16	4	20	"	"
P209	3B-3Y0	"	Open	17,35, 36	15	37	3	17	"	"
P210	3B-3Y1	"	35	17,36	"	18	"	"	"	"
P211	3B-3Y4	"	36	17,35	"	19	"	"	"	"
P212	3B-3Y5	"	35,36	17	"	40	"	"	"	"
P213	3B-3Y2	"	Open	17,35, 36	"	38	4	20	"	"
P214	3B-3Y3	"	35	17,36	"	39	"	"	"	"
P215	3B-3Y6	"	36	17,35	"	20	"	"	"	"
P216	3B-3Y7	"	35,36	17	"	16	"	"	"	"
P217	3A-3Y0	"	Open	15,17, 36	35	37	3	17	"	"
P218	3A-3Y2	"	15	17,36	"	38	"	"	"	"
P219	3A-3Y4	"	36	15,17	"	19	"	"	"	"
P220	3A-3Y6	"	15,36	17	"	20	"	"	"	"
P221	3A-3Y1	"	Open	15,17, 36	"	18	4	20	"	"
P222	3A-3Y3	"	15	17,36	"	39	"	"	"	"
P223	3A-3Y5	"	36	15,17	"	40	"	"	"	"
P224	3A-3Y7	"	15,36	17	"	16	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	1G1-1Y0	Propagation delay time (t_{PHL}), C_L = 50 pF	Open	3,22, 23,24	2	7	2	15	ns	16,17
Q2	1G1-1Y1	"	24	3,22, 23	"	27	"	"	"	"
Q3	1G1-1Y2	"	3	22,23, 24	"	6	"	"	"	"
Q4	1G1-1Y3	"	3,24	22,23	"	26	"	"	"	"
Q5	1G1-1Y4	"	23	3,22, 24	"	5	"	"	"	"
Q6	1G1-1Y5	"	23,24	3,22	"	25	"	"	"	"
Q7	1G1-1Y6	"	3,23	22,24	"	4	"	"	"	"
Q8	1G1-1Y7	"	3,23, 24	22	"	21	"	"	"	"
Q9	2G1-2Y0	"	Open	8,9, 28,31	30	11	"	"	"	"
Q10	2G1-2Y1	"	28	8,9,31	"	32	"	"	"	"
Q11	2G1-2Y2	"	8	9,28, 31	"	34	"	"	"	"
Q12	2G1-2Y3	"	8,28	9,31	"	14	"	"	"	"
Q13	2G1-2Y4	"	9	8,28, 31	"	13	"	"	"	"
Q14	2G1-2Y5	"	9,28	8,31	"	33	"	"	"	"
Q15	2G1-2Y6	"	9,8	28,31	"	12	"	"	"	"
Q16	2G1-2Y7	"	8,9,28	31	"	29	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q17	1G1-1Y0 Propagation delay time (t_{PHL}), C_L = 150 pF	Open	3,22, 23,24	2	7	5	20	ns	16,17
Q18	1G1-1Y1 "	24	3,22, 23	"	27	"	"	"	"
Q19	1G1-1Y2 "	3	22,23, 24	"	6	"	"	"	"
Q20	1G1-1Y3 "	3,24	22,23	"	26	"	"	"	"
Q21	1G1-1Y4 "	23	3,22, 24	"	5	"	"	"	"
Q22	1G1-1Y5 "	23,24	3,22	"	25	"	"	"	"
Q23	1G1-1Y6 "	3,23	22,24	"	4	"	"	"	"
Q24	1G1-1Y7 "	3,23, 24	22	"	21	"	"	"	"
Q25	2G1-2Y0 "	Open	8,9, 28,31	30	11	"	"	"	"
Q26	2G1-2Y1 "	28	8,9,31	"	32	"	"	"	"
Q27	2G1-2Y2 "	8	9,28, 31	"	34	"	"	"	"
Q28	2G1-2Y3 "	8,28	9,31	"	14	"	"	"	"
Q29	2G1-2Y4 "	9	8,28, 31	"	13	"	"	"	"
Q30	2G1-2Y5 "	9,28	8,31	"	33	"	"	"	"
Q31	2G1-2Y6 "	8,9	28,31	"	12	"	"	"	"
Q32	2G1-2Y7 "	8,9,28	31	"	29	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q33	1G2-1Y0	Propagation delay time (t_{PHL}), C_L = 50 pF	2	3,23, 24	22	7	2	15	ns	16,17
Q34	1G2-1Y1	"	2,24	3,23	"	27	"	"	"	"
Q35	1G2-1Y2	"	2,3	23,24	"	6	"	"	"	"
Q36	1G2-1Y3	"	2,3,24	23	"	26	"	"	"	"
Q37	1G2-1Y4	"	2,23	3,24	"	5	"	"	"	"
Q38	1G2-1Y5	"	2,23, 24	3	"	25	"	"	"	"
Q39	1G2-1Y6	"	2,3,23	24	"	4	"	"	"	"
Q40	1G2-1Y7	"	2,3, 23,24	Open	"	21	"	"	"	"
Q41	2G2-2Y0	"	30	8,9,28	31	11	"	"	"	"
Q42	2G2-2Y1	"	28,30	8,9	"	32	"	"	"	"
Q43	2G2-2Y2	"	8,30	9,28	"	34	"	"	"	"
Q44	2G2-2Y3	"	8,28, 30	9	"	14	"	"	"	"
Q45	2G2-2Y4	"	9,30	8,28	"	13	"	"	"	"
Q46	2G2-2Y5	"	9,28, 30	8	"	33	"	"	"	"
Q47	2G2-2Y6	"	8,9,30	28	"	12	"	"	"	"
Q48	2G2-2Y7	"	8,9, 28,30	Open	"	29	"	"	"	"
Q49	3G2-3Y0	"	Open	15,35, 36	17	37	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q50	3G2-3Y1	Propagation delay time (t_{PHL}), C_L = 50 pF	35	15,36	17	18	2	15	ns	16,17
Q51	3G2-3Y2	"	15	35,36	"	38	"	"	"	"
Q52	3G2-3Y3	"	15,35	36	"	39	"	"	"	"
Q53	3G2-3Y4	"	36	15,36	"	19	"	"	"	"
Q54	3G2-3Y5	"	35,36	15	"	40	"	"	"	"
Q55	3G2-3Y6	"	15,36	35	"	20	"	"	"	"
Q56	3G2-3Y7	"	15,35, 36	Open	"	16	"	"	"	"
Q57	1G2-1Y0	Propagation delay time (t_{PHL}), C_L = 150 pF	2	3,23, 24	22	7	5	20	"	"
Q58	1G2-1Y1	"	2,24	3,23	"	27	"	"	"	"
Q59	1G2-1Y2	"	2,3	23,24	"	6	"	"	"	"
Q60	1G2-1Y3	"	2,3,24	23	"	26	"	"	"	"
Q61	1G2-1Y4	"	2,23	3,24	"	5	"	"	"	"
Q62	1G2-1Y5	"	2,23, 24	3	"	25	"	"	"	"
Q63	1G2-1Y6	"	2,3,23	24	"	4	"	"	"	"
Q64	1G2-1Y7	"	2,3, 23,24	Open	"	21	"	"	"	"
Q65	2G2-2Y0	"	30	8,9,28	31	11	"	"	"	"
Q66	2G2-2Y1	"	28,30	8,9	31	32	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q67	2G2-2Y2	Propagation delay time (t_{PHL}), C_L = 150 pF	8,30	9,28	31	34	5	20	ns	16,17
Q68	2G2-2Y3	"	8,28, 30	9	"	14	"	"	"	"
Q69	2G2-2Y4	"	9,30	8,28	"	13	"	"	"	"
Q70	2G2-2Y5	"	9,28, 30	8	31	33	"	"	"	"
Q71	2G2-2Y6	"	8,9,30	28	"	12	"	"	"	"
Q72	2G2-2Y7	"	8,9, 28,30	Open	"	29	"	"	"	"
Q73	3G2-3Y0	"	Open	15,35, 36	17	37	"	"	"	"
Q74	3G2-3Y1	"	35	15,36	"	18	"	"	"	"
Q75	3G2-3Y2	"	15	35,36	"	38	"	"	"	"
Q76	3G2-3Y3	"	15,35	36	"	39	"	"	"	"
Q77	3G2-3Y4	"	36	15,35	"	19	"	"	"	"
Q78	3G2-3Y5	"	35,36	15	"	40	"	"	"	"
Q79	3G2-3Y6	"	15,36	35	"	20	"	"	"	"
Q80	3G2-3Y7	"	15,35, 36	Open	"	16	"	"	"	"
Q81	1C-1Y0	Propagation delay time (t_{PHL}), C_L = 50 pF	2	3,22, 24	23	7	2	15	"	"
Q82	1G1-1Y1	Propagation delay time (t_{PHL}), C_L = 50 pF	2,24	3,22	23	27	2	15	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q83	1C-1Y2	Propagation delay time (t_{PHL}), C_L = 50 pF	2,3	22,24	23	6	2	15	ns	16,17
Q84	1C-1Y3	"	2,3,24	22	"	26	"	15	"	"
Q85	1C-1Y4	"	2	3,22, 24	"	5	"	16	"	"
Q86	1C-1Y5	"	2,24	3,22	"	25	"	"	"	"
Q87	1C-1Y6	"	2,3	22,24	"	4	"	"	"	"
Q88	1C-1Y7	"	2,3,24	22	"	21	"	"	"	"
Q89	1B-1Y0	"	2	22,23, 24	3	7	"	15	"	"
Q90	1B-1Y1	"	2,24	22,23	"	27	"	"	"	"
Q91	1B-1Y4	"	2,23	22,24	"	5	"	"	"	"
Q92	1B-1Y5	"	2,23, 24	22	"	25	"	"	"	"
Q93	1B-1Y2	"	2	22,23, 24	"	6	"	16	"	"
Q94	1B-1Y3	"	2,24	22,23	"	26	"	"	"	"
Q95	1B-1Y6	"	2,23	22,24	"	4	"	"	"	"
Q96	1B-1Y7	"	2,23, 24	22	"	21	"	"	"	"
Q97	1A-1Y0	"	2	3,22, 23	24	7	"	15	"	"
Q98	1A-1Y2	"	2,3	22,23	24	6	"	15	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q99	1A-1Y4	Propagation delay time (t_{PHL}), C_L = 50 pF	2,23	3,22	24	5	2	15	ns	16,17
Q100	1A-1Y6	"	2,3,23	22	"	4	"	15	"	"
Q101	1A-1Y1	"	2	3,22, 23	"	27	"	16	"	"
Q102	1A-1Y3	"	2,3	22,23	"	26	"	"	"	"
Q103	1A-1Y5	"	2,23	3,22	"	25	"	"	"	"
Q104	1A-1Y7	"	2,3,23	22	"	21	"	"	"	"
Q105	2C-2Y0	"	30	8,28, 31	9	11	"	15	"	"
Q106	2C-2Y1	"	28,30	8,31	"	32	"	"	"	"
Q107	2C-2Y2	"	8,30	28,31	"	34	"	"	"	"
Q108	2C-2Y3	"	8,28, 30	31	"	14	"	"	"	"
Q109	2C-2Y4	"	30	8,28, 31	"	13	"	16	"	"
Q110	2C-2Y5	"	28,30	8,31	"	33	"	"	"	"
Q111	2C-2Y6	"	8,30	28,31	"	12	"	"	"	"
Q112	2C-2Y7	"	8,28, 30	31	"	29	"	"	"	"
Q113	2B-2Y0	"	30	9,28, 31	8	11	"	15	"	"
Q114	2B-2Y1	"	28,30	9,31	8	32	"	15	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q115	2B-2Y4	Propagation delay time (t_{PHL}), C_L = 50 pF	9,30	28,31	8	13	2	15	ns	16,17
Q116	2B-2Y5	"	9,28, 30	31	"	33	"	15	"	"
Q117	2B-2Y2	"	30	9,28, 31	"	34	"	16	"	"
Q118	2B-2Y3	"	28,30	9,31	"	14	"	"	"	"
Q119	2B-2Y6	"	9,30	28,31	"	12	"	"	"	"
Q120	2B-2Y7	"	9,28, 30	31	"	29	"	"	"	"
Q121	2A-2Y0	"	30	8,9,31	28	11	"	15	"	"
Q122	2A-2Y2	"	8,30	9,31	"	34	"	"	"	"
Q123	2A-2Y4	"	9,30	8,31	"	13	"	"	"	"
Q124	2A-2Y6	"	8,9,30	31	"	12	"	"	"	"
Q125	2A-2Y1	"	30	8,9,31	"	32	"	16	"	"
Q126	2A-2Y3	"	8,30	9,31	"	14	"	"	"	"
Q127	2A-2Y5	"	9,30	8,31	"	33	"	"	"	"
Q128	2A-2Y7	"	8,9,30	31	"	29	"	"	"	"
Q129	3C-3Y0	"	Open	15,17, 35	36	37	"	15	"	"
Q130	3C-3Y1	"	35	15,17	"	18	"	"	"	"
Q131	3C-3Y2	"	15	17,35	"	38	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q132	3C-3Y3	Propagation delay time (t_{PHL}), C_L = 50 pF	15,35	17	36	39	2	15	ns	16,17
Q133	3C-3Y4	"	Open	15,17, 35	"	19	"	16	"	"
Q134	3C-3Y5	"	35	15,17	"	40	"	"	"	"
Q135	3C-3Y6	"	15	17,35	"	20	"	"	"	"
Q136	3C-3Y7	"	15,35	17	"	16	"	"	"	"
Q137	3B-3Y0	"	Open	17,35, 36	15	37	"	15	"	"
Q138	3B-3Y1	"	35	17,36	"	18	"	"	"	"
Q139	3B-3Y4	"	36	17,35	"	19	"	"	"	"
Q140	3B-3Y5	"	35,36	17	"	40	"	"	"	"
Q141	3B-3Y2	"	Open	17,35, 36	"	38	"	16	"	"
Q142	3B-3Y3	"	35	17,36	"	39	"	"	"	"
Q143	3B-3Y6	"	36	17,35	"	20	"	"	"	"
Q144	3B-3Y7	"	35,36	17	"	16	"	"	"	"
Q145	3A-3Y0	"	Open	15,17, 36	35	37	"	15	"	"
Q146	3A-3Y2	"	15	17,36	"	38	"	"	"	"
Q147	3A-3Y4	"	36	15,17	"	19	"	"	"	"
Q148	3A-3Y6	"	15,36	17	"	20	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q149	3A-3Y1	Propagation delay time (t_{PHL}), C_L = 50 pF	Open	15,17, 36	35	18	2	16	ns	16,17
Q150	3A-3Y3	"	15	17,36	"	39	"	"	"	"
Q151	3A-3Y5	"	36	15,17	"	40	"	"	"	"
Q152	3A-3Y7	"	15,36	17	"	16	"	"	"	"
Q153	1C-1Y0	Propagation delay time (t_{PHL}), C_L = 150 pF	2	3,22, 24	23	7	5	20	"	"
Q154	1C-1Y1	"	2,24	3,22	"	27	"	"	"	"
Q155	1C-1Y2	"	2,3	22,24	"	6	"	"	"	"
Q156	1C-1Y3	"	2,3,24	22	"	26	"	"	"	"
Q157	1C-1Y4	"	2	3,22, 24	"	5	"	21	"	"
Q158	1C-1Y5	"	2,24	3,22	"	25	"	"	"	"
Q159	1C-1Y6	"	2,3	22,24	"	4	"	"	"	"
Q160	1C-1Y7	"	2,3,24	22	"	21	"	"	"	"
Q161	1B-1Y0	"	2	22,23, 24	3	7	"	20	"	"
Q162	1B-1Y1	"	2,24	22,23	"	27	"	"	"	"
Q163	1B-1Y4	"	2,3	22,24	"	5	"	"	"	"
Q164	1B-1Y5	"	2,3,24	22	"	25	"	"	"	"
Q165	1B-1Y2	"	2	22,23, 24	"	6	"	21	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q166	1B-1Y3	Propagation delay time (t_{PHL}), C_L = 150 pF	2,24	22,23	3	26	5	21	ns	16,17
Q167	1B-1Y6	"	2,3	22,24	"	4	"	"	"	"
Q168	1B-1Y7	"	2,3,24	22	"	21	"	"	"	"
Q169	1A-1Y0	"	2	3,22, 23	24	7	"	20	"	"
Q170	1A-1Y2	"	2,3	22,23	"	6	"	"	"	"
Q171	1A-1Y4	"	2,23	3,22	"	5	"	"	"	"
Q172	1A-1Y6	"	2,3,23	22	"	4	"	"	"	"
Q173	1A-1Y1	"	2	3,22, 23	"	27	"	21	"	"
Q174	1A-1Y3	"	2,3	22,23	"	26	"	"	"	"
Q175	1A-1Y5	"	2,23	3,22	"	25	"	"	"	"
Q176	1A-1Y7	"	2,3,23	22	"	21	"	"	"	"
Q177	2C-2Y0	"	30	8,28, 31	9	11	"	20	"	"
Q178	2C-2Y1	"	28,30	8,31	"	32	"	"	"	"
Q179	2C-2Y2	"	8,30	28,31	"	34	"	"	"	"
Q180	2C-2Y3	"	8,28, 30	31	"	14	"	"	"	"
Q181	2C-2Y4	"	30	8,28, 31	"	13	"	21	"	"
Q182	2C-2Y5	"	28,30	8,31	"	33	"	21	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q183	2C-2Y6	Propagation delay time (t_{PHL}), C_L = 150 pF	8,30	28,31	9	12	5	21	ns	16,17
Q184	2C-2Y7	"	8,28, 30	31	9	29	"	21	"	"
Q185	2B-2Y0	"	30	9,28, 31	8	11	"	20	"	"
Q186	2B-2Y1	"	28,30	9,31	"	32	"	"	"	"
Q187	2B-2Y4	"	9,30	28,31	"	13	"	"	"	"
Q188	2B-2Y5	"	9,28, 30	31	"	33	"	"	"	"
Q189	2B-2Y2	"	30	9,28, 31	"	34	"	21	"	"
Q190	2B-2Y3	"	28,30	9,31	"	14	"	"	"	"
Q191	2B-2Y6	"	9,30	28,31	"	12	"	"	"	"
Q192	2B-2Y7	"	9,28, 30	31	"	29	"	"	"	"
Q193	2A-2Y0	"	30	8,9,31	28	11	"	20	"	"
Q194	2A-2Y2	"	8,30	9,31	"	34	"	"	"	"
Q195	2A-2Y4	"	9,30	8,31	"	13	"	"	"	"
Q196	2A-2Y6	"	8,9,30	31	"	12	"	"	"	"
Q197	2A-2Y1	"	30	8,9,31	"	32	"	21	"	"
Q198	2A-2Y3	"	8,30	9,31	"	14	"	"	"	"
Q199	2A-2Y5	"	9,30	8,31	"	33	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

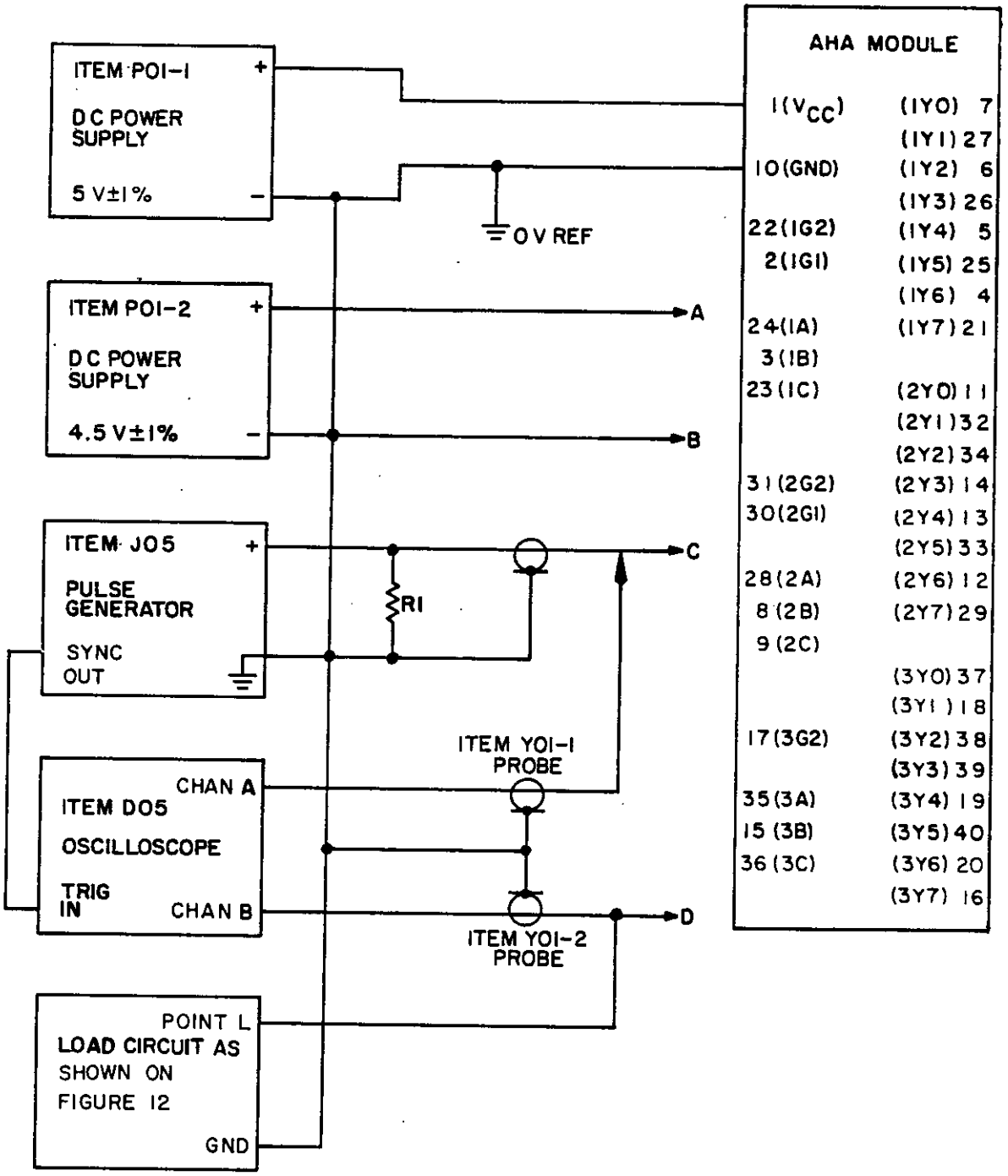
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q200	2A-2Y7	Propagation delay time (t_{PHL}), C_L = 150 pF	8,9,30	31	28	29	5	21	ns	16,17
Q201	3C-3Y0	"	Open	15,17, 35	36	37	"	20	"	"
Q202	3C-3Y1	"	35	15,17	"	18	"	"	"	"
Q203	3C-3Y2	"	15	17,35	"	38	"	"	"	"
Q204	3C-3Y3	"	15,35	17	"	39	"	"	"	"
Q205	3C-3Y4	"	Open	15,17, 35	"	19	"	21	"	"
Q206	3C-3Y5	"	35	15,17	"	40	"	"	"	"
Q207	3C-3Y6	"	15	17,35	"	20	"	"	"	"
Q208	3C-3Y7	"	15,35	17	"	16	"	"	"	"
Q209	3B-3Y0	"	Open	17,35, 36	15	37	"	20	"	"
Q210	3B-3Y1	"	35	17,36	"	18	"	"	"	"
Q211	3B-3Y4	"	36	17,35	"	19	"	"	"	"
Q212	3B-3Y5	"	35,36	17	"	40	"	"	"	"
Q213	3B-3Y2	"	Open	17,35, 36	"	38	"	21	"	"
Q214	3B-3Y3	"	35	17,36	"	39	"	"	"	"
Q215	3B-3Y6	"	36	17,35	"	20	"	"	"	"
Q216	3B-3Y7	"	35,36	17	"	16	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q217	3A-3Y0	Propagation delay time (t_{PHL}), C_L = 150 pF	Open	15,17, 36	35	37	5	20	ns	16,17
Q218	3A-3Y2	"	15	17,36	"	38	"	"	"	"
Q219	3A-3Y4	"	36	15,17	"	19	"	"	"	"
Q220	3A-3Y6	"	15,36	17	"	20	"	"	"	"
Q221	3A-3Y1	"	Open	15,17, 36	"	18	"	21	"	"
Q222	3A-3Y3	"	15	17,36	"	39	"	"	"	"
Q223	3A-3Y5	"	36	15,17	"	40	"	"	"	"
Q224	3A-3Y7	"	15,36	17	"	16	"	"	"	"

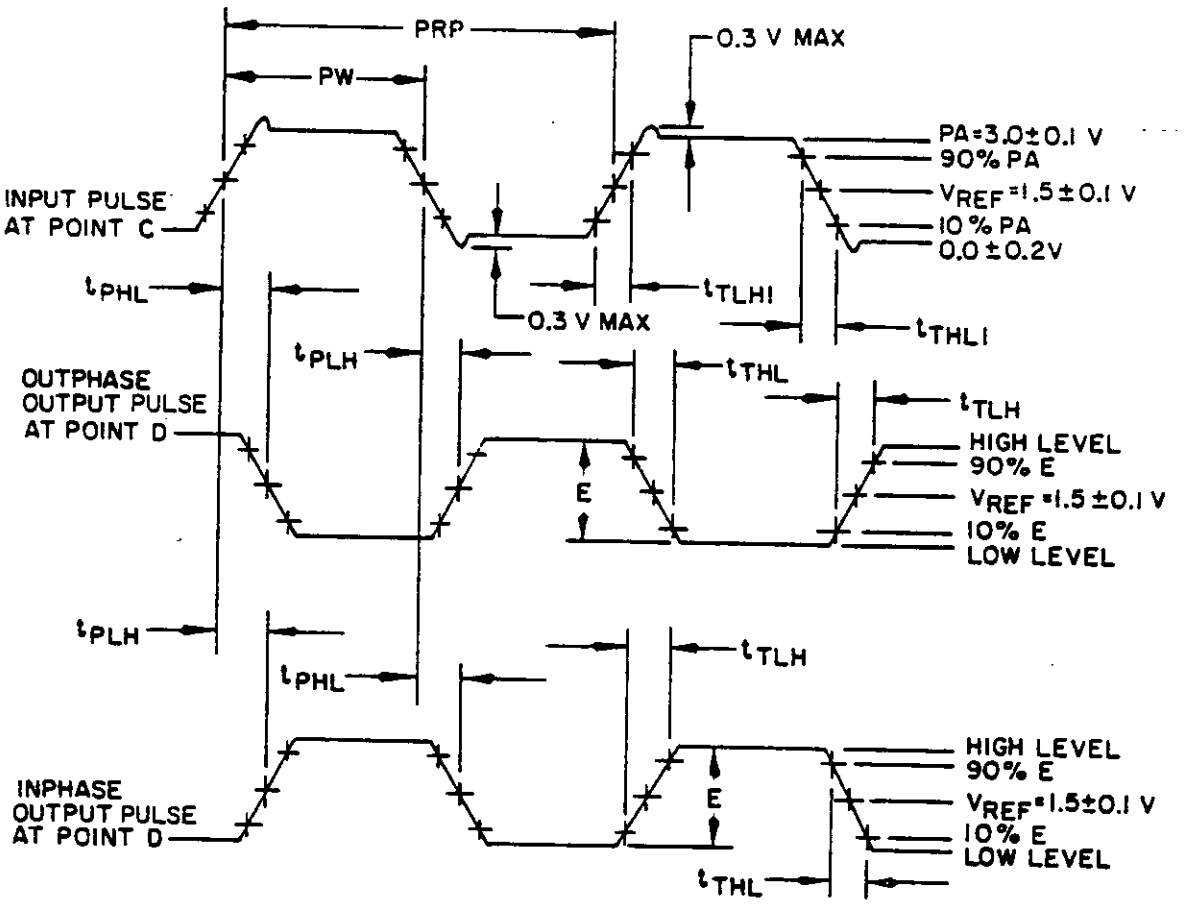
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NOTE: R1 = 51Ω ±5%, 1/2 W.

* FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XIV.

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NOTES:
1. Input pulse parameters:
   PW = 100 ±10 ns.
   PRP = 500 ±50 ns.
   $t_{TLH1} = t_{THL1} = 4 \pm 2$ ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. All voltages are referenced to ground.

* FIGURE 17. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XIV.

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TABLE XV. Output crosstalk (low-level and high-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1Y0	Output crosstalk, low level	2,30, 35	3,17, 22,23, 24,31	8,9,15, 28,36	7	0	500	mV	18,19
R2	1Y1	"	2,24, 28,30	3,8,17, 22,23, 31,36	9,15, 35	27	"	"	"	"
R3	1Y2	"	2,3,30, 35	17,22, 23,24, 31	8,9,15, 28,36	6	"	"	"	"
R4	1Y3	"	2,3,24, 28,30	8,17, 22,23, 31,36	9,15, 35	26	"	"	"	"
R5	1Y4	"	2,23, 30,35	3,17, 22,24, 31	8,9,15, 28,36	5	"	"	"	"
R6	1Y5	"	2,23, 24,30, 35	3,17, 22,31	8,9,15, 28,36	25	"	"	"	"
R7	1Y6	"	2,3,23, 30,35	17,22, 24,31	8,9,15, 28,36	4	"	"	"	"
R8	1Y7	"	2,3,23, 24,28, 30	8,17, 22,31, 36	9,15, 35	21	"	"	"	"
R9	2Y0	"	2,30, 35	8,9,17, 22,23, 24,28, 31	3,15,36	11	"	"	"	"
R10	2Y1	"	2,28, 30,24	8,9,17, 22,23, 31,36	3,15,35	32	"	"	"	"

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TABLE XV. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R11	2Y2	Output crosstalk, low level	2,8,30, 24	9,17, 22,23, 28,31, 36	3,15,35	34	0	500	mV	18,19
R12	2Y3	"	2,8,28, 30,35	9,17, 22,23, 24,31	3,15,36	14	"	"	"	"
R13	2Y4	"	2,9,30, 35	8,17, 22,23, 24,28, 31	3,15, 36	13	"	"	"	"
R14	2Y5	"	2,9,24, 28,30	8,17, 22,23, 31,36	3,15,35	33	"	"	"	"
R15	2Y6	"	2,8,9, 30,35	17,22, 23,24, 28,31	3,15,36	12	"	"	"	"
R16	2Y7	"	2,8,9, 28,30, 35	17,22, 23,24, 31	3,15,36	29	"	"	"	"
R17	3Y0	"	2,8,24, 30	9,15, 17,22, 23,31 35,36	3,28	37	"	"	"	"
R18	3Y1	"	2,30,35	9,15, 17,22, 23,24, 31,36	3,8,28	18	"	"	"	"
R19	3Y2	"	2,15,30	9,17, 22,23, 24,31, 35,36	3,8,28	38	"	"	"	"

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TABLE XV. Output crosstalk (low-level and high-level) test - Continued.

Test condition		Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R20	3Y3	Output crosstalk, low level	2,8,15, 24,30, 35	9,17, 22,23, 31,36	3,28	39	0	500	mV	18,19
R21	3Y4	"	2,30,36	9,15, 17,22, 23,24, 31,35	3,8,28	19	"	"	"	"
R22	3Y5	"	2,8,24, 30,35, 36	9,15, 17,22, 23,31	3,28	40	"	"	"	"
R23	3Y6	"	2,15, 30,36	9,17, 22,23, 24,31, 35	3,8,28	20	"	"	"	"
R24	3Y7	"	2,15, 30,35, 36	9,17, 22,23, 24,31	3,8,28	16	"	"	"	"
S1	1Y0	Output crosstalk, high level	22,30, 35	2,17,31	8,9,15, 28,36	7	"	400	"	"
S2	1Y1	"	22,28, 30	2,8,17, 31,36	9,15,35	27	"	"	"	"
S3	1Y2	"	22,30, 35	2,17, 31	8,9,15, 28,36	6	"	"	"	"
S4	1Y3	"	22,28, 30	2,8,17, 31,36	9,15,35	26	"	"	"	"
S5	1Y4	"	22,30, 35	2,17,31	8,9,15, 28,36	5	"	"	"	"
S6	1Y5	"	22,30, 35	2,17,31	8,9,15, 28,36	25	"	"	"	"

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TABLE XV. Output crosstalk (low-level and high-level) test - Continued.

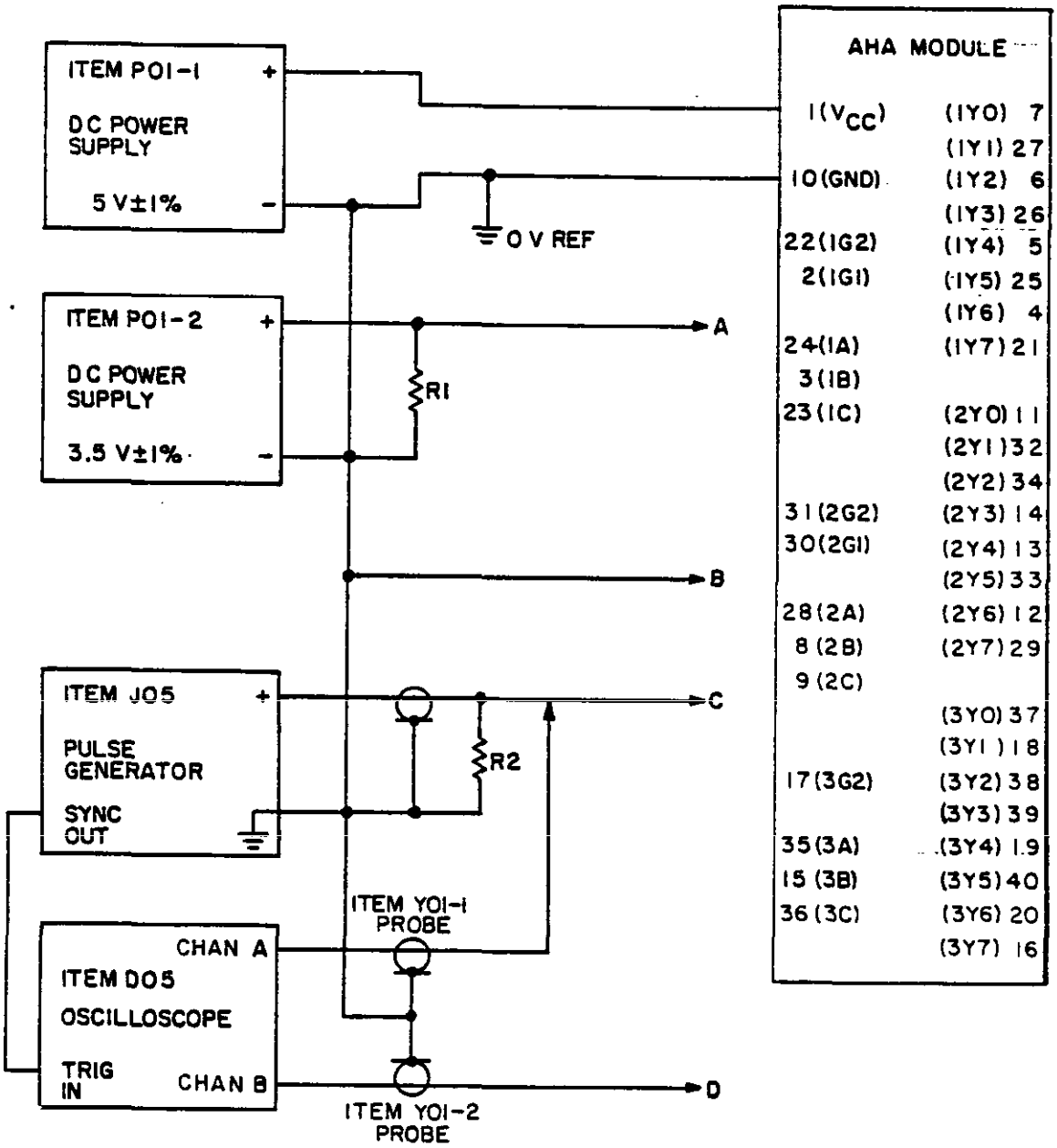
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S7	1Y6	Output crosstalk, high level	22,30, 35	2,17,31	8,9,15, 28,36	4	0	400	mV	18,19
S8	1Y7	"	22,28, 30	2,8,17, 31,36	9,15,35	21	"	"	"	"
S9	2Y0	"	2,31,35	17,22, 23,24, 30	3,15,36	11	"	"	"	"
S10	2Y1	"	2,24,31	17,22, 23,30, 36	3,15,35	32	"	"	"	"
S11	2Y2	"	2,24,31	17,22, 23,30, 36	3,15,35	34	"	"	"	"
S12	2Y3	"	2,31,35	17,22, 23,24, 30	3,15,36	14	"	"	"	"
S13	2Y4	"	2,31,35	17,22, 23,24, 30	3,15,36	13	"	"	"	"
S14	2Y5	"	2,24,31	17,22, 23,30, 36	3,15,35	33	"	"	"	"
S15	2Y6	"	2,31,35	17,22, 23,24, 30	3,15,36	12	"	"	"	"
S16	2Y7	"	2,31,35	17,22, 23,24, 30	3,15,36	29	"	"	"	"

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TABLE XV. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S17	3Y0	Output crosstalk, high level	2,8,17, 24,30	9,22, 23,31	3,28	37	0	400	mV	18,19
S18	3Y1	"	2,17,30	9,22, 23,24, 31	3,8,28	18	"	"	"	"
S19	3Y2	"	2,17,30	9,22, 23,24, 31	3,8,28	38	"	"	"	"
S20	3Y3	"	2,8,17, 24,30	9,22, 23,31	3,28	39	"	"	"	"
S21	3Y4	"	2,17,30	9,22, 23,24, 31	3,8,28	19	"	"	"	"
S22	3Y5	"	2,8,17, 24,30	9,22, 23,31	3,28	40	"	"	"	"
S23	3Y6	"	2,17,30	9,22, 23,24, 31	3,8,28	20	"	"	"	"
S24	3Y7	"	2,8,17, 24,30	9,22, 23,31	3,28	16	"	"	"	"

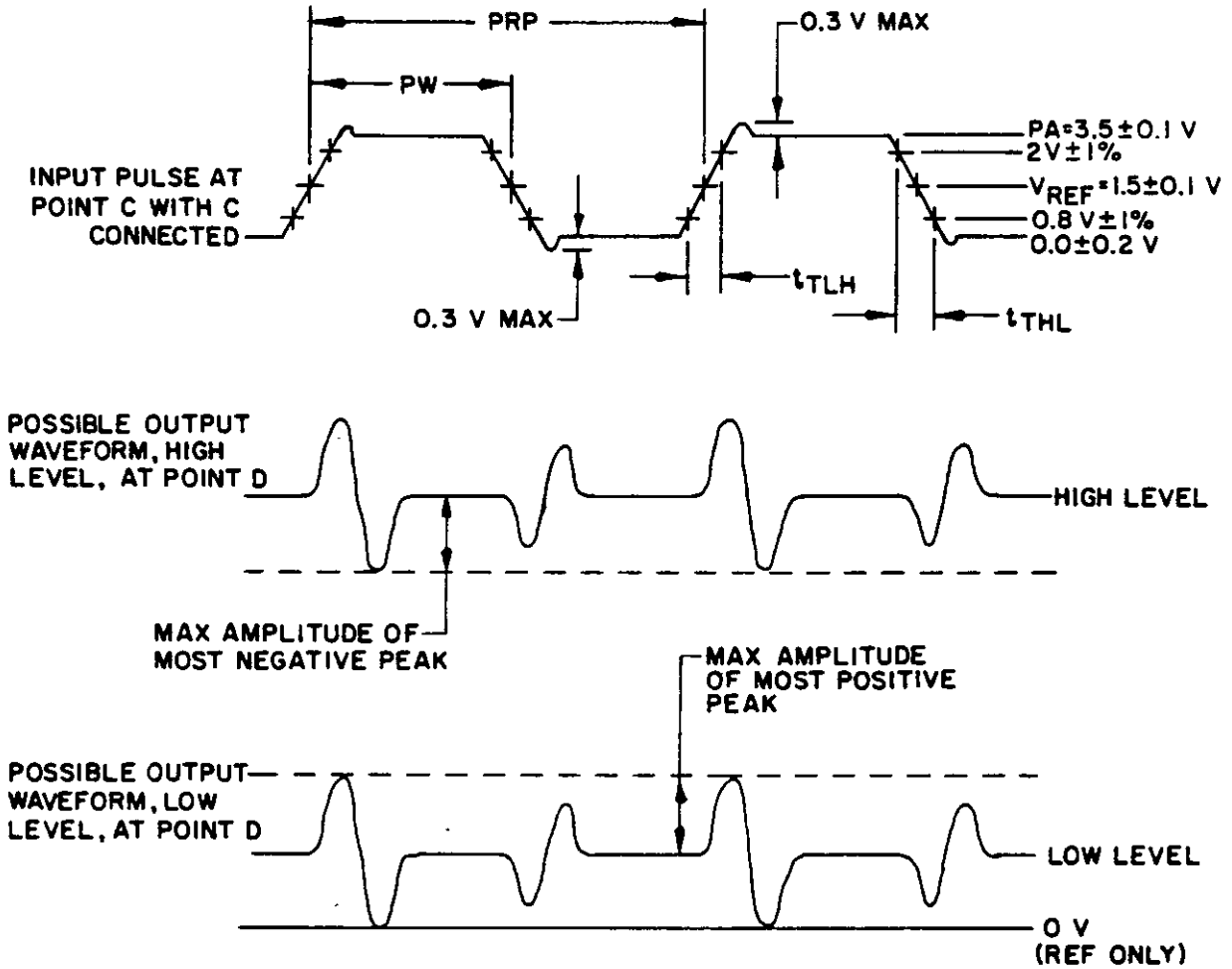
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs as shown on figure 12. C_L = 50 pF for test conditions R1-R24 and S1-S24.

FIGURE 18. Output crosstalk (low-level and high-level) test circuit for table XV.

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NOTE: Input pulse parameters:
$t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at all input pins).
PW = 250 ±25 ns.
PRP = 500 ±50 ns.

FIGURE 19. Output crosstalk (high-level and low-level) test circuit waveforms for table XV.

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TABLE XVI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
T	Power supply transient amplitude	0	400	mV	20,21

TABLE XVII. Power supply filter capacitance test.

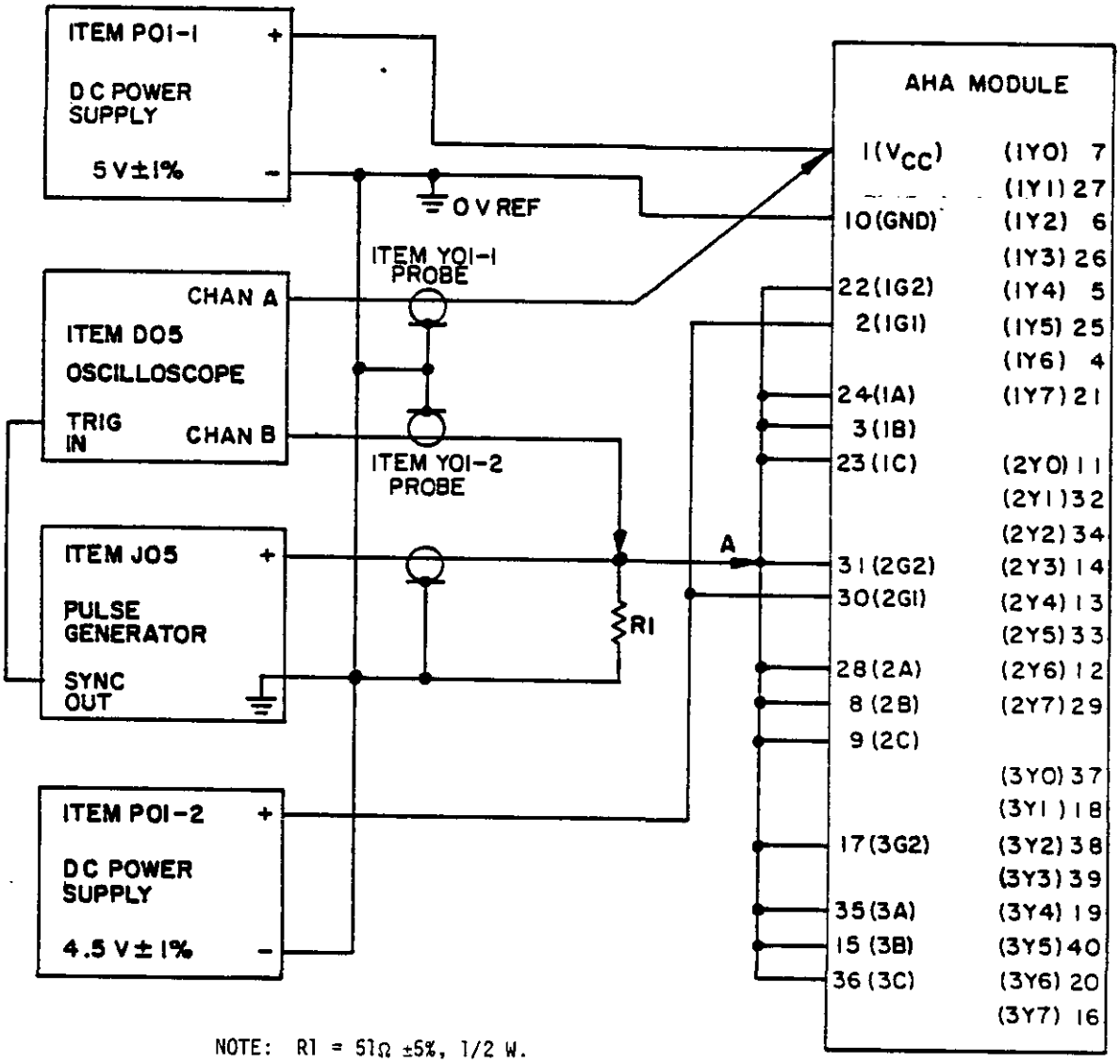
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U	Power supply filter capacitance	Open	1	1.6	3.0	µF	22

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TABLE XVIII. Input capacitance test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
V1	1G2	Input capacitance	22	Open	5	30	pF	22
V2	1G1	"	2	"	"	15	"	"
V3	1A	"	24	"	"	"	"	"
V4	1B	"	3	"	"	"	"	"
V5	1C	"	23	"	"	"	"	"
V6	2G2	"	31	"	"	30	"	"
V7	2G1	"	30	"	"	15	"	"
V8	2A	"	28	"	"	"	"	"
V9	2B	"	8	"	"	"	"	"
V10	2C	"	9	"	"	"	"	"
V11	3G2	"	17	"	"	30	"	"
V12	3A	"	35	"	"	15	"	"
V13	3B	"	15	"	"	"	"	"
V14	3C	"	36	"	"	"	"	"

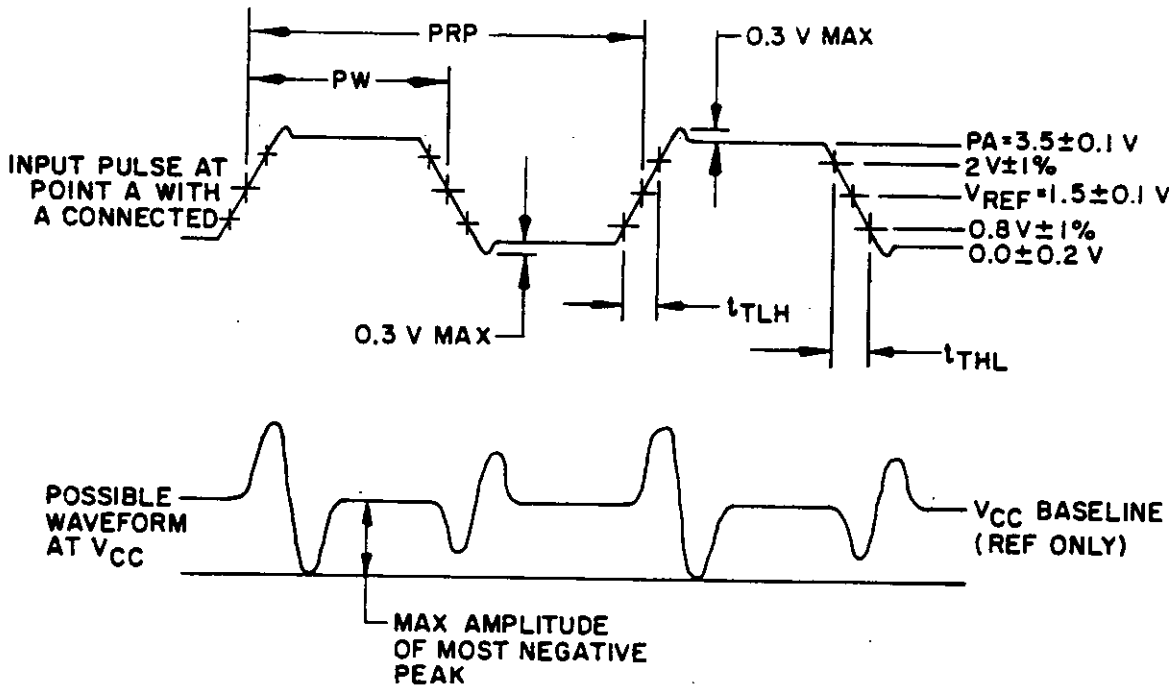
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NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 20. Power supply transient amplitude test circuit for table XVI.

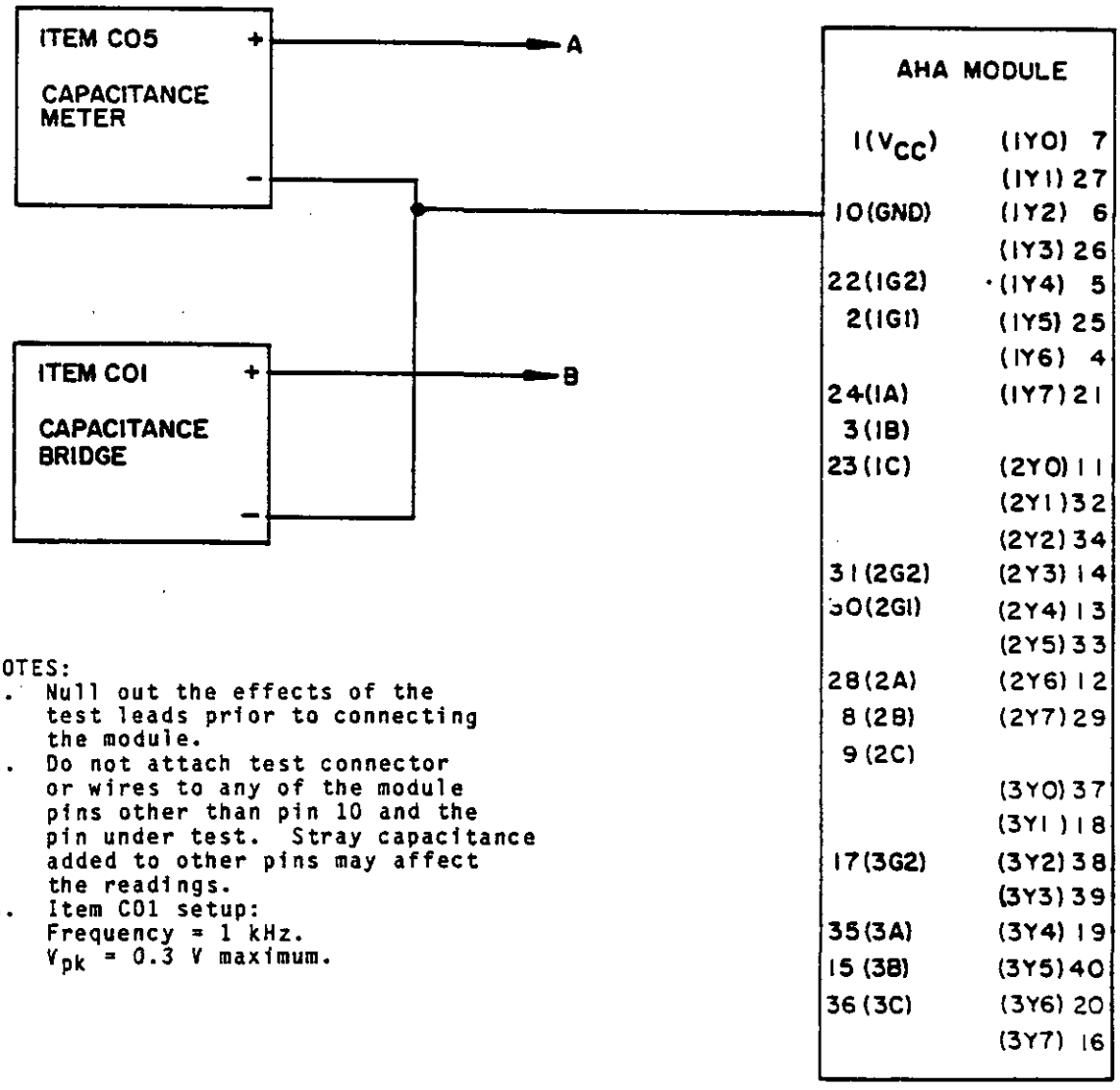
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NOTE: Input parameters:
$t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at each input pin).
PW = 250 ±25 ns.
PRP = 500 ±50 ns.

FIGURE 21. Power supply transient amplitude test circuit waveforms for table XVI.

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item CO1 setup:
   Frequency = 1 kHz.
   V_{pk} = 0.3 V maximum.

* FIGURE 22. Power supply filter and input capacitance test circuit for tables XVII and XVIII.

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TABLE XIX. Functional test.

Test condition	Parameter	Connect			Output logic states 1/																								Test figure
		A to pin	B to pin	C to pin	1Y0	1Y1	1Y2	1Y3	1Y4	1Y5	1Y6	1Y7	2Y0	2Y1	2Y2	2Y3	2Y4	2Y5	2Y6	2Y7	3Y0	3Y1	3Y2	3Y3	3Y4	3Y5	3Y6	3Y7	
W1	Functional test	2,17,31	3,22,23, 24	2/ 3/	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	23
W2	"	2,17, 24,31	3,22,23	"	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
M3	"	2,3,17, 31	22,23,24	"	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W4	"	2,3,17, 24,31	22,23	"	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W5	"	2,17, 23,31	3,22,24	"	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W6	"	2,17,23, 24,31	3,22	"	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W7	"	2,3,17, 23,31	22,24	"	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W8	"	2,3,17, 23,24,31	22	"	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W9	"	17,22,30	8,9,28,31	"	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W10	"	17,22, 28,30	8,9,31	"	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W11	"	8,17, 22,30	9,28,31	"	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W12	"	8,17,22, 28,30	9,31	"	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	"
W13	"	9,17,22, 30	8,28,31	"	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	"

See footnotes at end of table.

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TABLE XIX. Functional test - Continued.

Test condition	Parameter	Connect			Output logic states 1/																								Test figure
		A to pin	B to pin	C to pin	1Y0	1Y1	1Y2	1Y3	1Y4	1Y5	1Y6	1Y7	2Y0	2Y1	2Y2	2Y3	2Y4	2Y5	2Y6	2Y7	3Y0	3Y1	3Y2	3Y3	3Y4	3Y5	3Y6	3Y7	
W14	Functional test	9,17,22, 28,30	8,31	2/ 3/	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	23
W15	"	8,9,17, 22,30	28,31	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	"
W16	"	8,9,17, 22,28,30	31	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	"
W17	"	22,31	15,17, 35,36	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	"
W18	"	22,31,35	15,17,36	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	"
W19	"	22,31,15	17,35,36	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	"
W20	"	15,22, 31,35	17,36	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	"
W21	"	22,31,36	15,17,35	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	"
W22	"	22,31, 35,36	15,17	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	"
W23	"	15,22, 31,36	17,35	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	"
W24	"	15,22,31, 35,36	17	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	"
W25	"	2,17,22, 30,31	3,8,9,15, 23,24,28, 35,36	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W26	"	2,17,22, 24,28,30, 31,35	3,8,9,15, 23,36	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"

See footnotes at end of table.

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TABLE XIX. Functional test - Continued.

Test condition	Parameter	Connect			Output logic states 1/																								Test figure
		A to pin	B to pin	C to pin	1Y0	1Y1	1Y2	1Y3	1Y4	1Y5	1Y6	1Y7	2Y0	2Y1	2Y2	2Y3	2Y4	2Y5	2Y6	2Y7	3Y0	3Y1	3Y2	3Y3	3Y4	3Y5	3Y6	3Y7	
W27	Functional test	2,3,8,15, 17,22,30, 31	9,23,24, 28,35,36	2/ 3/	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	23
W28	"	2,3,8,15, 17,22,24, 28,30,31, 35	9,23,36	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W29	"	2,9,17, 22,23,30, 31,36	3,8,15, 24,28,35	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W30	"	2,9,17, 22,23,24, 28,30,31, 35,36	3,8,15	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W31	"	2,3,8,9, 15,17,22, 23,30,31, 36	24,28,35	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W32	"	2,3,8,9, 15,17,22, 23,24,28, 30,31,35, 36	Open	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W33	"	15,17, 35,36	2,3,8,9, 22,23,24, 28,30,31	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W34	"	15,17,24, 28,35,36	2,3,8,9, 22,23,30, 31	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"

See footnotes at end of table.

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TABLE XIX. Functional test - Continued.

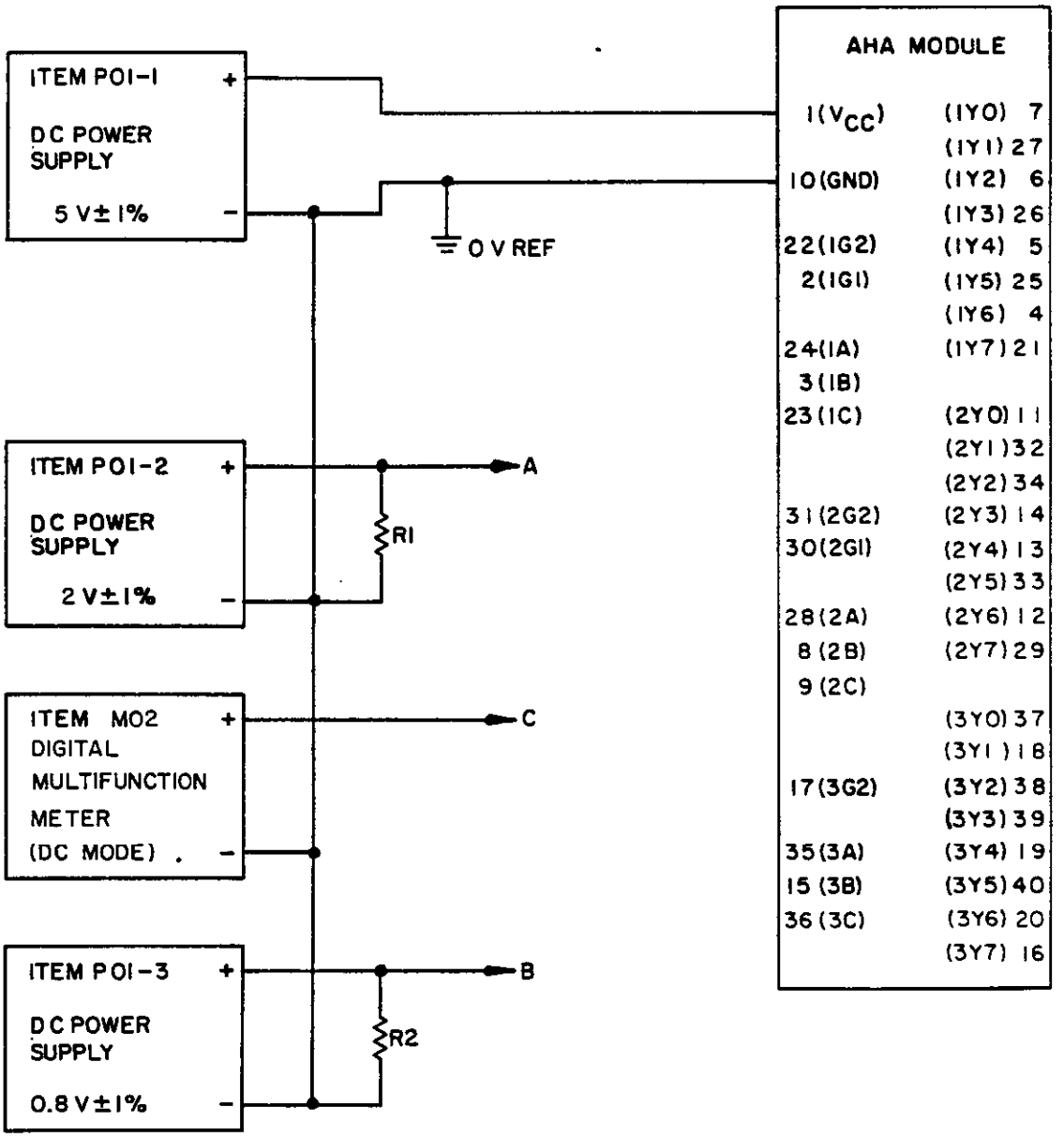
Test condition	Parameter	Connect			Output logic states 1/																								Test figure
		A to pin	B to pin	C to pin	1Y0	1Y1	1Y2	1Y3	1Y4	1Y5	1Y6	1Y7	2Y0	2Y1	2Y2	2Y3	2Y4	2Y5	2Y6	2Y7	3Y0	3Y1	3Y2	3Y3	3Y4	3Y5	3Y6	3Y7	
W35	Functional test	3,8,15, 17,35,36	2,9,22, 23,24,28, 30,31	2/ 3/	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	23
W36	"	3,8,15, 17,24,28, 35,36	2,9,22, 23,30,31	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W37	"	9,15,17, 23,35,36	2,3,8, 22,24,28, 30,31	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W38	"	9,15,17, 23,24,28, 35,36	2,3,8, 22,30,31	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W39	"	3,8,9,15, 17,23,35, 36	2,22,24, 28,30,31	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
W40	"	3,8,9,15, 17,23,24, 28,35,36	2,22, 30,31	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"

1/ Output logic states:
L = Low level = 0.0 V to 0.5 V.
H = High level = 2.5 V to 4.5 V.

2/ For test conditions W1-W40, verification of the output logic states in footnote 1 is obtained by connecting item M02 (point C) to each output pin in succession after input conditions have been accomplished.

3/ 1Y0-1Y7, 2Y0-2Y7, and 3Y0-3Y7 output logic states are measured at output pins 7,27,6,26,5,25,4,21,11,32,34,14,13,33,12,29,37,18,38,39,19,40,20, and 16, respectively.

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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

*FIGURE 23. Functional test circuit for table XIX.

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